

VECTOR 3D

JEE EXERCISES

Exercise J-1

BASIC PROPERTIES OF VECTORS

1. The vertices of a triangle are A(1, 1, 2), B(4, 3, 1) and C(2, 3, 5). A vector representing the internal bisector of the angle A is :

एक त्रिभुज के शीर्ष A(1, 1, 2), B(4, 3, 1) और C(2, 3, 5) हैं, तो कोण A के अन्तः समद्विभाजक को निरूपित करने वाला सदिश है:

- (A) $\hat{i} + \hat{j} + 2\hat{k}$ (B) $2\hat{i} - 2\hat{j} + \hat{k}$ (C) $2\hat{i} + 2\hat{j} - \hat{k}$ (D) $2\hat{i} + 2\hat{j} + \hat{k}$

Ans. D

Sol. D is mid-point of AB

D(3,3,3)

$$\overrightarrow{AD} = 2\hat{i} + 2\hat{j} + \hat{k}$$

2. Consider points A, B, C and D with position vectors $7\hat{i} - 4\hat{j} + 7\hat{k}$, $\hat{i} - 6\hat{j} + 10\hat{k}$, $-\hat{i} - 3\hat{j} + 4\hat{k}$ and $5\hat{i} - \hat{j} + 5\hat{k}$ respectively. Then ABCD is a :

- (A) parallelogram but not a rhombus (B) square
(C) rhombus (D) None of these

माना बिन्दु A, B, C तथा D हैं जिनके स्थिति सदिश क्रमशः $7\hat{i} - 4\hat{j} + 7\hat{k}$, $\hat{i} - 6\hat{j} + 10\hat{k}$, $-\hat{i} - 3\hat{j} + 4\hat{k}$ तथा $5\hat{i} - \hat{j} + 5\hat{k}$ हैं, तब ABCD होगा :

- (A) समान्तर चतुर्भुज परन्तु समचतुर्भुज नहीं (B) वर्ग
(C) समचतुर्भुज (D) इनमें से कोई नहीं

Ans. D

Sol. AB = 7, BC = 7

$$CD = \sqrt{41}$$

$$AD = \sqrt{17}$$

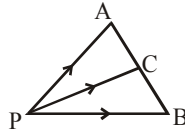
3. If C is the mid point of AB and P is any point outside AB, then :

यदि C, AB का मध्य बिन्दु P है तथा कोई भी एक बिन्दु है, जो AB के बाहर है, तो :

- (A) $\overrightarrow{PA} + \overrightarrow{PB} = 2\overrightarrow{PC}$ (B) $\overrightarrow{PA} + \overrightarrow{PB} = \overrightarrow{PC}$
(C) $\overrightarrow{PA} + \overrightarrow{PB} + 2\overrightarrow{PC} = \vec{0}$ (D) $\overrightarrow{PA} + \overrightarrow{PB} + \overrightarrow{PC} = \vec{0}$

Ans. A

Sol. $\overrightarrow{PA} + \overrightarrow{PB} = (\overrightarrow{PA} + \overrightarrow{AC}) + (\overrightarrow{PB} + \overrightarrow{BC}) - (\overrightarrow{AC} + \overrightarrow{BC})$
 $= \overrightarrow{PC} + \overrightarrow{PC} - (\overrightarrow{AC} - \overrightarrow{CB})$
 $= 2\overrightarrow{PC} - 0$
 $(\because \overrightarrow{AC} = \overrightarrow{CB})$
 $\therefore \overrightarrow{PA} + \overrightarrow{PB} = 2\overrightarrow{PC}$



4. Let $\vec{a}, \vec{b}, \vec{c}$ be three non-zero vectors which are pairwise non-collinear. If $\vec{a} + 3\vec{b}$ is collinear with $\vec{c}$ and $\vec{b} + 2\vec{c}$ is collinear with $\vec{a}$, then $\vec{a} + 3\vec{b} + 6\vec{c}$ is : [AIEEE-2011]

माना $\vec{a}, \vec{b}, \vec{c}$ तीन शून्येतर सदिश हैं जो युग्मत सरेख नहीं हैं। यदि $\vec{a} + 3\vec{b}$ तथा $\vec{c}$ सरेख हैं तथा $\vec{b} + 2\vec{c}$ तथा $\vec{a}$ सरेख हैं, तो $\vec{a} + 3\vec{b} + 6\vec{c}$ बराबर हैं : [AIEEE-2011]

- (A) $\vec{a} + \vec{c}$ (B) $\vec{a}$ (C) $\vec{c}$ (D) $\vec{0}$

Ans. D

Sol. Let $\vec{a} + 3\vec{b} = \lambda \vec{c}$ Let $\vec{b} + 2\vec{c} = \mu \vec{a}$

add $6\vec{c}$ both side $3\vec{b} + 6\vec{c} = 3\mu \vec{a}$

$\vec{a} + 3\vec{b} + 6\vec{c} = (\lambda + 6)\vec{c}$ add $\vec{a}$ both side

$\vec{a} + 3\vec{b} + 6\vec{c} = (3\mu + 1)\vec{a}$

Hence $(\lambda + 6)\vec{c} = (3\mu + 1)\vec{a}$

But given $\vec{a}$ and $\vec{c}$ are non coliner

Hence $\lambda + 6 = 3\mu + 1 = 0$

so $\vec{a} + 3\vec{b} + 6\vec{c} = \vec{0}$

DOT PRODUCT & ITS APPLICATION

5. If $|\vec{a}| = 5$, $|\vec{a} - \vec{b}| = 8$ and $|\vec{a} + \vec{b}| = 10$, then $|\vec{b}|$ is equal to :

- (A) 1 (B) $\sqrt{57}$ (C) 3 (D) none of these

यदि $|\vec{a}| = 5$, $|\vec{a} - \vec{b}| = 8$ और $|\vec{a} + \vec{b}| = 10$ हो, तो $|\vec{b}| =$

- (A) 1 (B) $\sqrt{57}$ (C) 3 (D) इनमें से कोई नहीं

Ans. B

Sol. $|\vec{a} - \vec{b}|^2 + |\vec{a} + \vec{b}|^2 = 164$

$$|\vec{a}|^2 + |\vec{b}|^2 = 82$$

$$25 + |\vec{b}|^2 = 82$$

$$|\vec{b}| = \sqrt{57}$$

6. Angle between diagonals of a parallelogram whose side are represented by $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} - \hat{k}$:

एक समान्तर चतुर्भुज जिसकी भुजाएँ $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$ तथा $\vec{b} = \hat{i} - \hat{j} - \hat{k}$ से दी गई है, के विकर्णों के मध्य कोण होगा :

- (A) $\cos^{-1}\left(\frac{1}{3}\right)$ (B) $\cos^{-1}\left(\frac{1}{2}\right)$ (C) $\cos^{-1}\left(\frac{4}{9}\right)$ (D) $\cos^{-1}\left(\frac{5}{9}\right)$

Ans. A

Sol. $\vec{\alpha} + (1, -1, -1) = (2, 1, 1)$

$$\vec{\alpha} = (1, 2, 2)$$

$$\vec{\beta} = (3, 0, 0)$$

$$\vec{\alpha} \cdot \vec{\beta} = 3 \times 3 \cos \theta = 3$$

$$\cos \theta = \frac{1}{3}$$

7. Two adjacent sides of a parallelogram ABCD are given by $\vec{AB} = 2\hat{i} + 10\hat{j} + 11\hat{k}$ and $\vec{AD} = -\hat{i} + 2\hat{j} + 2\hat{k}$. The side AD is rotated by an acute angle α in the plane of the parallelogram so that AD becomes AD'. If AD' makes a right angle with the side AB, then the cosine of the angle α is given by :

समान्तर चतुर्भुज ABCD की दो आसन्न भुजाएँ $\vec{AB} = 2\hat{i} + 10\hat{j} + 11\hat{k}$ तथा $\vec{AD} = -\hat{i} + 2\hat{j} + 2\hat{k}$ द्वारा दी गई है। यदि भुजा AD को न्यून कोण α द्वारा समान्तर चतुर्भुज के तल में घूर्णन किया जाता है। इस प्रकार कि यह AD' हो जाती है एवं AD' भुजा AB के साथ समकोण बनाती है तो कोण α की कोज्या होगी :

- (A) $\frac{8}{9}$ (B) $\frac{\sqrt{17}}{9}$ (C) $\frac{1}{9}$ (D) $\frac{4\sqrt{5}}{9}$

Ans. B

Sol. $\cos \theta = -2 + 20 + 22 = 40$

$$\cos \theta > 0 \Rightarrow \theta < 90$$

$$\theta + \alpha = 90$$

$$\cos \alpha = \cos(90 - \theta) = \sin \theta$$

$$= \frac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|}$$

$$\frac{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 10 & 11 \\ -1 & 2 & 2 \end{vmatrix}}{15 \times 3} = \frac{\hat{i}(-2) - \hat{j}(15) + \hat{k}(14)}{45}$$

$$= \frac{\sqrt{425}}{45} = \frac{\sqrt{17}}{9}$$

8. Let two non-collinear unit vector $\hat{a}$ and $\hat{b}$ form an acute angle. A point P moves so that at any time t the position vector $\vec{OP}$ (where O is the origin) is given by $\hat{a} \cos t + \hat{b} \sin t$. When P is farthest from origin O, let M be the length of $\vec{OP}$ and $\hat{u}$ be the unit vector along $\vec{OP}$. Then :

(A) $\hat{u} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|}$ and $M = (1 + \hat{a} \cdot \hat{b})^{\frac{1}{2}}$

(B) $\hat{u} = \frac{\hat{a} - \hat{b}}{|\hat{a} - \hat{b}|}$ and $M = (1 + \hat{a} \cdot \hat{b})^{\frac{1}{2}}$

(C) $\hat{u} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|}$ and $M = (1 + 2\hat{a} \cdot \hat{b})^{\frac{1}{2}}$

(D) $\hat{u} = \frac{\hat{a} - \hat{b}}{|\hat{a} - \hat{b}|}$ and $M = (1 + 2\hat{a} \cdot \hat{b})^{\frac{1}{2}}$

माना असंरेखीय इकाई सदिश $\hat{a}$ तथा $\hat{b}$ एक न्यूनकोण बनाते हैं। एक बिन्दु P ऐसे गमन करता है कि किसी समय t पर स्थिति सदिश $\vec{OP}$ (जहाँ O मूल बिन्दु है) $\hat{a} \cos t + \hat{b} \sin t$ द्वारा दिया जाता है। माना जब P मूल बिन्दु O से अधिकतम दूरी पर है, तब $\vec{OP}$ की लम्बाई M तथा $\vec{OP}$ के समान्तर इकाई सदिश $\hat{u}$ है, तब :

(A) $\hat{u} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|}$ तथा $M = (1 + \hat{a} \cdot \hat{b})^{\frac{1}{2}}$

(B) $\hat{u} = \frac{\hat{a} - \hat{b}}{|\hat{a} - \hat{b}|}$ तथा $M = (1 + \hat{a} \cdot \hat{b})^{\frac{1}{2}}$

(C) $\hat{u} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|}$ तथा $M = (1 + 2\hat{a} \cdot \hat{b})^{\frac{1}{2}}$

(D) $\hat{u} = \frac{\hat{a} - \hat{b}}{|\hat{a} - \hat{b}|}$ तथा $M = (1 + 2\hat{a} \cdot \hat{b})^{\frac{1}{2}}$

Ans. A

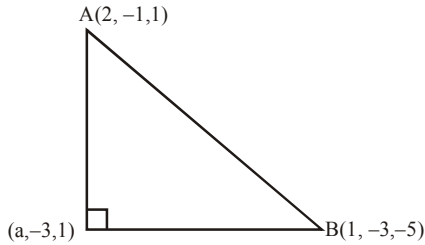
Sol. $AC \perp BC$

$\therefore$ dr's of AC and BC will be $(2-a, 2, 0)$ and $(1-a, 0, -6)$

So that $(2-a)(1-a) + 2 \times 0 + 0 \times (-6) = 0$

$$\Rightarrow a^2 - 3a + 2 = 0$$

$$\therefore a = 1, 2$$



9. Let $\hat{a}$ and $\hat{b}$ be two unit vectors. If the vectors $\vec{c} = \hat{a} + 2\hat{b}$ and $\vec{d} = 5\hat{a} - 4\hat{b}$ are perpendicular to each other, then the angle between $\hat{a}$ and $\hat{b}$ is : [AIEEE-2012]

माना $\hat{a}$ तथा $\hat{b}$ दो मात्रक सदिश हैं। यदि $\vec{c} = \hat{a} + 2\hat{b}$ तथा $\vec{d} = 5\hat{a} - 4\hat{b}$ परस्पर लम्बवत् हैं, तो $\hat{a}$ तथा $\hat{b}$ के बीच का कोण है : [AIEEE-2012]

- (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{3}$

Ans. D

Sol. $\vec{c} \cdot \vec{d} = 0$

$$\Rightarrow (\hat{a} + 2\hat{b}) \cdot (5\hat{a} - 4\hat{b}) = 0$$

$$\Rightarrow 5 - 8 + 6\hat{a} \cdot \hat{b} = 0$$

$$\Rightarrow \hat{a} \cdot \hat{b} = 1/2$$

$$\Rightarrow \cos \theta = 1/2$$

$$\Rightarrow \theta = \frac{\pi}{3}$$

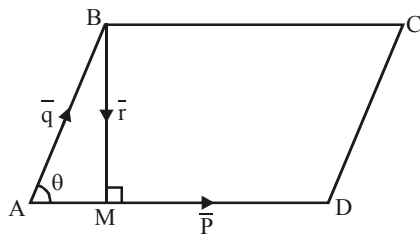
10. Let ABCD be a parallelogram such that $AB = \vec{q}$, $AD = \vec{p}$ and $\angle BAD$ be an acute angle. If $\vec{r}$ is the vector that coincides with the altitude directed from the vertex B to the side AD, then $\vec{r}$ is given by : [AIEEE-2012]

माना ABCD एक ऐसा समांतर चतुर्भुज है, कि $AB = \vec{q}$, $AD = \vec{p}$ तथा $\angle BAD$ एक न्यून कोण है। यदि सदिश $\vec{r}$ शीर्ष B से AD पर खींचे गए लंब के संपाती है, तो $\vec{r}$ निम्न द्वारा प्रदत्त है : [AIEEE-2012]

- (A) $\vec{r} = -3\vec{q} + \frac{3(\vec{p} \cdot \vec{q})}{(\vec{p} \cdot \vec{p})} \vec{p}$ (B) $\vec{r} = 3\vec{q} - \frac{3(\vec{p} \cdot \vec{q})}{(\vec{p} \cdot \vec{p})} \vec{p}$ (C) $\vec{r} = -\vec{q} + \left(\frac{\vec{p} \cdot \vec{q}}{\vec{p} \cdot \vec{p}} \right) \vec{p}$ (D) $\vec{r} = \vec{q} - \left(\frac{\vec{p} \cdot \vec{q}}{\vec{p} \cdot \vec{p}} \right) \vec{p}$

Ans. C

Sol.



$$\vec{q} + \vec{r} = \vec{AM}$$

$$\Rightarrow \vec{r} = -\vec{q} + \vec{AM}$$

$$\Rightarrow \vec{r} = -\vec{q} + \frac{\vec{p} \cdot \vec{q}}{|\vec{p}|^2} \vec{p}$$

$$\Rightarrow \vec{r} = -\vec{q} + \left(\frac{\vec{p} \cdot \vec{q}}{\vec{p} \cdot \vec{p}} \right) \vec{p}$$

11. Let $\vec{u}$ be a vector coplanar with the vectors $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{b} = \hat{j} + \hat{k}$. If $\vec{u}$ is perpendicular to $\vec{a}$ and $\vec{u} \cdot \vec{b} = 24$, then $|\vec{u}|^2$ is equal to : [JEE(Main)-2018]

माना $\vec{u}$ एक ऐसा सदिश है जो सदिशों $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$ तथा $\vec{b} = \hat{j} + \hat{k}$ के साथ समतलीय है। यदि $\vec{u}$, $\vec{a}$ पर लंबवत् है तथा $\vec{u} \cdot \vec{b} = 24$ है, तो $|\vec{u}|^2$ बराबर है : [JEE(Main)-2018]

- (A) 256 (B) 84 (C) 336 (D) 315

Ans. C

Sol. $\vec{u} = \lambda (\vec{a} \times (\vec{a} \times \vec{b}))$

$$= \lambda ((\vec{a} \cdot \vec{b})\vec{a} - (\vec{a} \cdot \vec{a})\vec{b})$$

$$= \lambda (2\vec{a} - 14\vec{b})$$

$$\vec{u} = \lambda (4\hat{i} - 8\hat{j} - 16\hat{k})$$

$$\vec{u} \cdot \vec{b} = 24 \Rightarrow \lambda = -1$$

$$\Rightarrow \vec{u} = -4\hat{i} + 8\hat{j} + 16\hat{k}$$

$$\Rightarrow |\vec{u}|^2 = 16 + 64 + 256 = 336$$

CROSS PRODUCT AND ITS APPLICATION

12. Let $\vec{a} = \hat{i} + \hat{j}$ and $\vec{b} = 2\hat{i} - \hat{k}$. The point of intersection of the lines $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$ and $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$ is :

माना कि $\vec{a} = \hat{i} + \hat{j}$ तथा $\vec{b} = 2\hat{i} - \hat{k}$ हैं, रेखाओं $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$ तथा $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$ का प्रतिच्छेद बिन्दु है :

- (A) $-\hat{i} + \hat{j} + 2\hat{k}$ (B) $3\hat{i} - \hat{j} + \hat{k}$ (C) $3\hat{i} + \hat{j} - \hat{k}$ (D) $\hat{i} - \hat{j} - \hat{k}$

Ans. C

Sol. $\vec{a} = \hat{i} + \hat{j}$ and $\vec{b} = 2\hat{i} - \hat{k}$

$$\vec{r} \times \vec{a} = \vec{b} \times \vec{a} \Rightarrow (\vec{r} - \vec{b}) \times \vec{a} = 0$$

$$\Rightarrow \vec{r} = \vec{b} + \lambda \vec{a} \quad \dots(i)$$

similarly $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$

$$\Rightarrow \vec{r} = \vec{a} + \mu \vec{b} \quad \dots(ii)$$

Putting the vector $\vec{a}$ and $\vec{b}$ in (i) and (ii) and equating

$$\text{we get } 2\hat{i} - \hat{k} + \lambda(\hat{i} + \hat{j}) = \hat{i} + \hat{j} + \mu(2\hat{i} - \hat{k})$$

$$\Rightarrow 2 + \lambda = 1 + 2\mu, \lambda = 1, \mu = 1$$

$\therefore$ Point of intersection is $3\hat{i} + \hat{j} - \hat{k}$.

13. Unit vector perpendicular to the plane of the triangle ABC with position vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ of the vertices A, B, C is : (where Δ is the area of the triangle ABC).

(A) $\frac{(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})}{\Delta}$

(B) $\frac{(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})}{2\Delta}$

(C) $\frac{(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})}{4\Delta}$

(D) None of these

त्रिभुज ABC के शीर्षों A, B, C के स्थिति सदिश क्रमशः $\vec{a}$, $\vec{b}$, $\vec{c}$ है, तो त्रिभुज के समतल के लम्बवत् इकाई सदिश है :
(जहाँ Δ त्रिभुज ABC का क्षेत्रफल है)

(A) $\frac{(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})}{\Delta}$

(B) $\frac{(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})}{2\Delta}$

(C) $\frac{(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})}{4\Delta}$

(D) इनमें से कोई नहीं

Ans. B

Sol. $\vec{\Delta} = |\vec{\Delta}| \hat{\Delta} = \frac{1}{2}(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})$

14. If $\vec{u}$ and $\vec{v}$ are unit vectors and θ is the acute angle between them, then $2\vec{u} \times 3\vec{v}$ is a unit vector for :

- (A) Exactly two values of θ (B) More than two values of θ
(C) No value of θ (D) Exactly one value of θ

यदि $\vec{u}$ तथा $\vec{v}$ इकाई सदिश हैं तथा θ उनके मध्य न्यून कोण है, तब $2\vec{u} \times 3\vec{v}$ जिनके लिए इकाई सदिश है :

- (A) θ के ठीक दो मानों के लिए (B) θ के दो मानों से ज्यादा के लिए
(C) θ के किसी मान के लिए नहीं (D) θ के ठीक एक मान के लिए

Ans. D

Sol. $|2\vec{u} \times 3\vec{v}| = 1$
 $|2\vec{u} \times 3\vec{v}| = 6|\vec{u}||\vec{v}|\sin\theta = 1$
 $= 6\sin\theta = 1$
 $\sin\theta = \frac{1}{6}$

15. Let $\vec{a}, \vec{b}, \vec{c}$ be unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. Which one of the following is correct?

- (A) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} = \vec{0}$ (B) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} \neq \vec{0}$
(C) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{a} \times \vec{c} \neq \vec{0}$ (D) $\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ are mutually perpendicular

माना कि $\vec{a}, \vec{b}, \vec{c}$ इकाई सदिश इस प्रकार हैं कि $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ निम्न में से कौनसा सत्य है?

- (A) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} = \vec{0}$ (B) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} \neq \vec{0}$
(C) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{a} \times \vec{c} \neq \vec{0}$ (D) $\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ परस्पर लम्बवत् है

Ans. B

Sol. $|\vec{a} + \vec{b} + \vec{c}| = 0$
 $\vec{b} \times \vec{a} + \vec{c} \times \vec{a} = 0$ $\vec{a} \times \vec{b} = \vec{c} \times \vec{a}$
 $\vec{a} \times \vec{b} + \vec{c} \times \vec{b} = 0$
 $\vec{a} \times \vec{c} + \vec{b} \times \vec{c} = 0$
 $\vec{a}, \vec{b}, \vec{c}$ are co-planar.
 $\vec{c} = -(\vec{a} + \vec{b})$

16. If the vectors $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} + \hat{k}$ and $\vec{c} = \lambda\hat{i} + \hat{j} + \mu\hat{k}$ are mutually orthogonal, then $(\lambda, \mu) =$ [AIEEE-2010]

यदि सदिश $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} + \hat{k}$ तथा $\vec{c} = \lambda\hat{i} + \hat{j} + \mu\hat{k}$ परस्पर लम्बकोणीय हैं, तो (λ, μ) बराबर है :

[AIEEE-2010]

- (A) $(-3, 2)$ (B) $(2, -3)$ (C) $(-2, 3)$ (D) $(3, -2)$

Ans. A

Sol. Give $\vec{a} \perp \vec{b}$, $\vec{a} \perp \vec{c}$ and $\vec{b} \perp \vec{c}$

$$\text{so } \vec{a} \cdot \vec{c} = 0 \text{ and } \vec{b} \cdot \vec{c} = 0$$

$$\Rightarrow \lambda - 1 + 2\mu = 0 \text{ and } 2\lambda + 4 + \mu = 0$$

$$\Rightarrow \lambda = -3 \text{ and } \mu = 2$$

17. The vectors $\vec{a}$ and $\vec{b}$ are not perpendicular and $\vec{c}$ and $\vec{d}$ are two vectors satisfying : $\vec{b} \times \vec{c} = \vec{b} \times \vec{d}$ and $\vec{a} \cdot \vec{d} = 0$. Then the vector $\vec{d}$ is equal to : [AIEEE-2011]

$\vec{a}$ तथा $\vec{b}$ लम्बवत् नहीं है तथा $\vec{c}$ तथा $\vec{d}$ इस प्रकार के हैं कि $\vec{b} \times \vec{c} = \vec{b} \times \vec{d}$ तथा $\vec{a} \cdot \vec{d} = 0$ हैं, तो $\vec{d}$ बराबर है :

[AIEEE-2011]

- (A) $\vec{b} + \left(\frac{\vec{b} \cdot \vec{c}}{\vec{a} \cdot \vec{b}} \right) \vec{c}$ (B) $\vec{c} - \left(\frac{\vec{a} \cdot \vec{c}}{\vec{a} \cdot \vec{b}} \right) \vec{b}$ (C) $\vec{b} - \left(\frac{\vec{b} \cdot \vec{c}}{\vec{a} \cdot \vec{b}} \right) \vec{c}$ (D) $\vec{c} + \left(\frac{\vec{a} \cdot \vec{c}}{\vec{a} \cdot \vec{b}} \right) \vec{b}$

Ans. B

Sol. $\vec{a} \cdot \vec{b} \neq 0$

$$\vec{a} \cdot \vec{d} = 0$$

$$\vec{b} \times \vec{c} = \vec{b} \times \vec{d}$$

$$\vec{a} \times (\vec{b} \times \vec{c}) = \vec{a} \times (\vec{b} \times \vec{d})$$

$$(\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} - (\vec{a} \cdot \vec{d})\vec{b} + (\vec{a} \cdot \vec{b})\vec{d} \quad \{\vec{a} \cdot \vec{d} = 0\}$$

$$\Rightarrow (\vec{a} \cdot \vec{b})\vec{d} = (\vec{a} \cdot \vec{b})\vec{c} \quad (\vec{a} \cdot \vec{b}) \text{ (divide by } \vec{a} \cdot \vec{b})$$

$$\boxed{\vec{d} = \vec{c} - \left(\frac{\vec{a} \cdot \vec{c}}{\vec{a} \cdot \vec{b}} \right) \vec{b}}$$

18. The value of $\left[(\vec{a} + 2\vec{b} - \vec{c}), (\vec{a} - \vec{b}), (\vec{a} - \vec{b} - \vec{c}) \right]$ is equal to the box product :

$\left[(\vec{a} + 2\vec{b} - \vec{c}), (\vec{a} - \vec{b}), (\vec{a} - \vec{b} - \vec{c}) \right]$ का मान निम्न में से किस बॉक्स गुणनफल (box product) के बराबर है:

- (A) $[\vec{a} \vec{b} \vec{c}]$ (B) $2[\vec{a} \vec{b} \vec{c}]$ (C) $3[\vec{a} \vec{b} \vec{c}]$ (D) $4[\vec{a} \vec{b} \vec{c}]$

Ans. C

Sol. $(\vec{a} + 2\vec{b} - \vec{c}), (\vec{a} - \vec{b}), (\vec{a} - \vec{b} - \vec{c})$
 $(\vec{a} - \vec{b}) \times (\vec{a} - \vec{b} - \vec{c})$
 $\vec{a} \times \vec{b} - \vec{a} \times \vec{c} - \vec{b} \times \vec{a} + \vec{b} \times \vec{c}$
 $(\vec{a} + 2\vec{b} - \vec{c}) \cdot (\vec{c} \times \vec{a} + \vec{b} \times \vec{a})$
 $2\vec{b} \cdot (\vec{c} \times \vec{a}) - \vec{c} \cdot (\vec{b} \times \vec{a})$
 $2[a, b, c] + [a, b, c] = 3[a, b, c]$

19. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + 2\hat{j} - \hat{k}$, then the value of $\begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$ is equal to :

यदि $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + 2\hat{j} - \hat{k}$ हो, तो $\begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$ का मान है :

- (A) 2 (B) 4 (C) 16 (D) 64

Ans. C

Sol. $\begin{vmatrix} 3 & 1 & 2 \\ 1 & 3 & -2 \\ 2 & -2 & 6 \end{vmatrix} = 3(18 - 4) - 1(6 + 4) + 2(-2 - 6)$

$= 42 - 10 - 16$
 $= 16$

20. If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar vectors and λ is a real number, then the vectors $\vec{a} + 2\vec{b} + 3\vec{c}$, $\lambda\vec{b} + 4\vec{c}$ and $(2\lambda - 1)\vec{c}$ are non-coplanar for :

- (A) all values of λ (B) all except one value of λ
 (C) all except two values of λ (D) no value of λ

यदि $\vec{a}, \vec{b}, \vec{c}$ विषमतलीय सदिश हैं तथा λ एक वास्तविक संख्या है तो सदिश $\vec{a} + 2\vec{b} + 3\vec{c}$, $\lambda\vec{b} + 4\vec{c}$ तथा $(2\lambda - 1)\vec{c}$ विषमतलीय है :

- (A) λ के सभी मानों के लिए (B) एक को छोड़ कर λ के सभी मानों के लिए
(C) दो को छोड़ कर λ के सभी मानों के लिए (D) λ के किसी भी मान के लिए नहीं

Ans. C

Sol. Let $\vec{\alpha} = \vec{a} + 2\vec{b} + 3\vec{c}$, $\vec{\beta} = \lambda\vec{b} + 4\vec{c}$ and $\vec{\gamma} = (2\lambda - 1)\vec{c}$.

$$\text{Then, } [\vec{\alpha} \vec{\beta} \vec{\gamma}] = \begin{vmatrix} 1 & 2 & 3 \\ 0 & \lambda & 4 \\ 0 & 0 & (2\lambda - 1) \end{vmatrix} [\vec{a} \vec{b} \vec{c}]$$

$$\Rightarrow [\vec{\alpha} \vec{\beta} \vec{\gamma}] = \lambda(2\lambda - 1)[\vec{a} \vec{b} \vec{c}]$$

$$\Rightarrow [\vec{\alpha} \vec{\beta} \vec{\gamma}] = 0, \text{ if } \lambda = 0, \frac{1}{2} \quad [\because [\vec{a} \vec{b} \vec{c}] \neq 0]$$

Hence, $\vec{\alpha}, \vec{\beta}, \vec{\gamma}$ are non-coplanar for all values of λ except two values 0 and $\frac{1}{2}$.

21. The edges of a parallelopiped are of unit length and are parallel to non-coplanar unit vectors $\hat{a}, \hat{b}, \hat{c}$ such that

$\hat{a} \cdot \hat{b} = \hat{b} \cdot \hat{c} = \hat{c} \cdot \hat{a} = \frac{1}{2}$. Then the volume of the parallelopiped is :

एक षट्फलक (parallelopiped) की भुजाएँ इकाई लम्बाई की है व विषमतलीय इकाई सदिशों $\hat{a}, \hat{b}, \hat{c}$ जहाँ $\hat{a} \cdot \hat{b} = \hat{b} \cdot \hat{c} = \hat{c} \cdot \hat{a} = \frac{1}{2}$ के समान्तर है। तो षट्फलक का आयतन निम्न है :

- (A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{\sqrt{3}}$

Ans. A

Sol. $\hat{a} \cdot \hat{b} = \hat{b} \cdot \hat{c} = \hat{c} \cdot \hat{a} = \frac{1}{2}$

$$V = (\vec{a} \times \vec{b}) \cdot \frac{\vec{a} \times \vec{b}}{\vec{a} \times \vec{b}} \cdot \vec{c} \quad h = \frac{(\hat{a} \times \hat{b})}{|\hat{a} \times \hat{b}|} \cdot \vec{c}$$

$$V = |(\vec{a} \times \vec{b}) \cdot \vec{c}|$$

$$[\vec{a} \quad \vec{b} \quad \vec{c}]^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix}$$

$$= 1 + \frac{1}{8} + \frac{1}{8} - \frac{1}{4} - \frac{1}{4} - \frac{1}{4}$$

$$= 1 - \frac{1}{2} = \frac{1}{2}$$

$$[\vec{a} \quad \vec{b} \quad \vec{c}] = \frac{1}{\sqrt{2}}$$

22. The vector $\vec{a} = \alpha \hat{i} + 2\hat{j} + \beta \hat{k}$ lies in the plane of the vectors $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \hat{j} + \hat{k}$ and bisects the angle between $\vec{b}$ and $\vec{c}$. Then which one of the following gives possible values of α and β ? [AIEEE-2008]

सदिश $\vec{a} = \alpha \hat{i} + 2\hat{j} + \beta \hat{k}$ सदिशों $\vec{b} = \hat{i} + \hat{j}$ तथा $\vec{c} = \hat{j} + \hat{k}$ के तल में स्थित है तथा $\vec{b}$ तथा $\vec{c}$ के बीच के कोण का समद्विभाजन करता है तो निम्न में से कौनसा α एवं β के संभावित मान प्रदान करता है ? [AIEEE-2008]

- (A) $\alpha = 1, \beta = 2$ (B) $\alpha = 2, \beta = 1$ (C) $\alpha = 1, \beta = 1$ (D) $\alpha = 2, \beta = 2$

Ans. C

Sol. $\vec{a} = \lambda(\hat{b} + \hat{c})$

$$\Rightarrow \alpha \hat{i} + 2\hat{j} + \beta \hat{k} = \lambda \left(\frac{\hat{i} + 2\hat{j} + \hat{k}}{\sqrt{2}} \right)$$

$$\lambda = \sqrt{2}\alpha \text{ and } \lambda = \sqrt{2} \text{ and } \lambda = \sqrt{2}\beta$$

$$\Rightarrow \alpha = 1 \text{ and } \beta = 1.$$

23. If $\vec{u}, \vec{v}, \vec{w}$ are non-coplanar vectors and p, q are real numbers, then the equality

$$[3\vec{u} \quad p\vec{v} \quad p\vec{w}] - [p\vec{v} \quad \vec{w} \quad q\vec{u}] - [2\vec{w} \quad q\vec{v} \quad q\vec{u}] = 0 \text{ holds for : [AIEEE-2009]}$$

- (A) exactly two values of (p, q)
 (C) all values of (p, q)

- (B) more than two but not all values of (p, q)
 (D) exactly one value of (p, q)

यदि $\vec{u}, \vec{v}, \vec{w}$ असमतलीय सदिश है तथा p, q वास्तविक संख्याएँ हैं तो समिका $[3\vec{u} \ \vec{p}\vec{v} \ \vec{p}\vec{w}] - [\vec{p}\vec{v} \ \vec{w} \ \vec{q}\vec{u}] - [2\vec{w} \ \vec{q}\vec{v} \ \vec{q}\vec{u}] = 0$

निम्न के लिए सत्य होगी :

[AIEEE-2009]

- (A) (p, q) के ठीक दो मानों के लिए
 (C) (p, q) के सभी मानों के लिए

- (B) दो से अधिक परन्तु (p, q) के सभी मानों के लिए नहीं
 (D) (p, q) के ठीक एक मान के लिए

Ans. D

Sol. $[3\vec{u} \ \vec{p}\vec{v} \ \vec{p}\vec{w}] - [\vec{p}\vec{v} \ \vec{w} \ \vec{q}\vec{u}] - [2\vec{w} \ \vec{q}\vec{v} \ \vec{q}\vec{u}] = 0$

$$3p^2[\vec{u} \ \vec{v} \ \vec{w}] - pq[\vec{v} \ \vec{w} \ \vec{u}] - 2q^2[\vec{w} \ \vec{v} \ \vec{u}] = 0$$

$$(3p^2 - pq + 2q^2) \cdot [\vec{u} \ \vec{v} \ \vec{w}] = 0$$

$$3p^2 - pq + 2q^2 = 0$$

has exactly one solution

$$p = q = 0$$

24. If the vectors $p\hat{i} + \hat{j} + \hat{k}, \hat{i} + q\hat{j} + \hat{k}$ and $\hat{i} + \hat{j} + r\hat{k}$ ($p \neq q \neq r \neq 1$) are coplanar, then the value of $pqr - (p + q + r)$ is : [AIEEE-2011]

यदि सदिश $p\hat{i} + \hat{j} + \hat{k}, \hat{i} + q\hat{j} + \hat{k}$ तथा $\hat{i} + \hat{j} + r\hat{k}$ ($p \neq q \neq r \neq 1$) समतलीय हैं, तो $pqr - (p + q + r)$ का मान है :

[AIEEE-2011]

- (A) -2 (B) 2 (C) 0 (D) -1

Ans. A

Sol.
$$\begin{vmatrix} p & 1 & 1 \\ 1 & q & 1 \\ 1 & 1 & r \end{vmatrix} = 0$$

$$p(qr - 1) - (r - 1) + (1 - q) = 0$$

$$pqr - p - r + 1 + 1 - q = 0$$

$$pqr - (p + r + q) + 2 = 0$$

$$pqr - (p + r + q) = -2$$

VTP AND ITS APPLICATION

25. If $\vec{b}$ and $\vec{c}$ are two non-collinear vectors such that $\vec{a} \parallel (\vec{b} \times \vec{c})$, then $(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{c})$ is equal to :

- (A) $\vec{a}^2 (\vec{b} \cdot \vec{c})$ (B) $\vec{b}^2 (\vec{a} \cdot \vec{c})$ (C) $\vec{c}^2 (\vec{a} \cdot \vec{b})$ (D) None of these

यदि $\vec{b}$ तथा $\vec{c}$ दो असंरेखीय सदिश इस प्रकार है कि $\vec{a} \parallel (\vec{b} \times \vec{c})$, तो $(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{c})$ बराबर होगा :

- (A) $\vec{a}^2 (\vec{b} \cdot \vec{c})$ (B) $\vec{b}^2 (\vec{a} \cdot \vec{c})$ (C) $\vec{c}^2 (\vec{a} \cdot \vec{b})$ (D) इनमें से कोई नहीं

Ans. A

Sol. $\vec{a} \parallel \vec{b} \times \vec{c}$

$$(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{c}) = ?$$

$$\vec{a} \perp \vec{b}$$

$$\vec{a} \perp \vec{c}$$

$$|\vec{a}| |\vec{b}| (\vec{a} \times \vec{b}) \cdot \vec{a} |\vec{c}| (\vec{a} \times \vec{c})$$

$$(\vec{a})^2 (|\vec{b}| |\vec{c}|) (\cos \theta)$$

$$\vec{a}^2 (\vec{b} \cdot \vec{c})$$

26. Vector of length 3 unit which is perpendicular to $\hat{i} + \hat{j} + \hat{k}$ and lies in the plane of $\hat{i} + \hat{j} + \hat{k}$ and $2\hat{i} - 3\hat{j}$:

परिमाण 3 का एक सदिश ज्ञात कीजिये जो $\hat{i} + \hat{j} + \hat{k}$ के लम्बवत् है तथा $\hat{i} + \hat{j} + \hat{k}$ तथा $2\hat{i} - 3\hat{j}$ के समतलीय है :

- (A) $\frac{3}{\sqrt{6}}(\hat{i} - 2\hat{j} + \hat{k})$ (B) $\frac{3}{\sqrt{6}}(2\hat{i} - \hat{j} - \hat{k})$ (C) $\frac{3}{\sqrt{114}}(8\hat{i} - 7\hat{j} - \hat{k})$ (D) $\frac{3}{\sqrt{114}}(-7\hat{i} + 8\hat{j} - \hat{k})$

Ans. D

Sol. $\vec{a} = \hat{i} + \hat{j} + \hat{k}$

$$\vec{b} = 2\hat{i} - 3\hat{j}$$

$$\vec{v} = (\vec{a} \times \vec{b}) \times \vec{a}$$

$$= (\vec{a} \cdot \vec{a}) \vec{b} - \vec{a} (\vec{a} \cdot \vec{b})$$

$$= 3(2\hat{i} - 3\hat{j}) - (-1)(\hat{i} + \hat{j} + \hat{k})$$

$$7\hat{i} - 8\hat{j} + \hat{k}$$

$$\text{For magnitude } 3 = \pm 3 \frac{\vec{v}}{|\vec{v}|}$$

$$= \pm \frac{3}{\sqrt{114}}(-7\hat{i} + 8\hat{j} - \hat{k})$$

27. If $[\vec{a} \vec{b} \vec{c}] = 4$ then $[\vec{a} \times \vec{b} \vec{b} \times \vec{c} \vec{c} \times \vec{a}] =$

यदि $[\vec{a} \vec{b} \vec{c}] = 4$ तब $[\vec{a} \times \vec{b} \vec{b} \times \vec{c} \vec{c} \times \vec{a}] =$

- (A) 4 (B) 2 (C) 8 (D) 16

Ans. D

Sol. $[a \ b \ c] = 4$

$$(\vec{a} \times \vec{b}) \times (\vec{b} \times \vec{c}) \cdot (\vec{c} \times \vec{a})$$

$$[\vec{b}((\vec{a} \times \vec{b}) \cdot \vec{c}) - \vec{c}((\vec{a} \times \vec{b}) \cdot \vec{b})] \cdot (\vec{c} \times \vec{a})$$

$$4\vec{b} \cdot (\vec{c} \times \vec{a})$$

$$= 16$$

28. If $\vec{c} = 2\lambda(\vec{a} \times \vec{b}) + 3\mu(\vec{b} \times \vec{a})$; $\vec{a} \times \vec{b} \neq 0$, $\vec{c} \cdot (\vec{a} \times \vec{b}) = 0$ then :

- (A) $\lambda = 3\mu$ (B) $2\lambda = 3\mu$ (C) $\lambda + \mu = 0$ (D) None of these

यदि $\vec{c} = 2\lambda(\vec{a} \times \vec{b}) + 3\mu(\vec{b} \times \vec{a})$; $\vec{a} \times \vec{b} \neq 0$, $\vec{c} \cdot (\vec{a} \times \vec{b}) = 0$ तो :

- (A) $\lambda = 3\mu$ (B) $2\lambda = 3\mu$ (C) $\lambda + \mu = 0$ (D) इनमें से कोई नहीं

Ans. B

Sol. $\vec{c} = 2\lambda(\vec{a} \times \vec{b}) + 3\mu(\vec{b} \times \vec{a})$ $\vec{a} \times \vec{b} \neq 0$

dot with $\vec{c}$ $\vec{c} \cdot (\vec{a} \times \vec{b}) = 0$

$$|\vec{c}|^2 = 0 + 0$$

$$\vec{c} = 0$$

$$-2\lambda + 3\mu = 0$$

29. For any vector $\vec{a}$, $|\vec{a} \times \hat{i}|^2 + |\vec{a} \times \hat{j}|^2 + |\vec{a} \times \hat{k}|^2$ is equal to :

- (A) $|\vec{a}|^2$ (B) $2|\vec{a}|^2$ (C) $3|\vec{a}|^2$ (D) None of these

किसी भी सदिश $\vec{a}$ के लिए, $|\vec{a} \times \hat{i}|^2 + |\vec{a} \times \hat{j}|^2 + |\vec{a} \times \hat{k}|^2$ का मान निम्न में से किस के बराबर है :

- (A) $|\vec{a}|^2$ (B) $2|\vec{a}|^2$ (C) $3|\vec{a}|^2$ (D) इनमें से कोई नहीं

Ans. B

Sol. $|\vec{a} \times \hat{i}|^2 + |\vec{a} \times \hat{j}|^2 + |\vec{a} \times \hat{k}|^2$
 $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$
 $(\vec{a} \times \hat{i} = -a_2\hat{k} + a_3\hat{j})$
 $=$
 $= 2(a_1^2 + a_2^2 + a_3^2)$
 $= 2|\vec{a}|^2$

$a_2^2 + a_3^2 + a_1^2 + a_3^2 + a_1^2 + a_2^2$

30. If $(\vec{a} \times \vec{b}) \times \vec{c} = \vec{a} \times (\vec{b} \times \vec{c})$, where $\vec{a}$, $\vec{b}$ and $\vec{c}$ are any three vectors such that $\vec{a} \cdot \vec{b} \neq 0$, $\vec{b} \cdot \vec{c} \neq 0$, then $\vec{a}$ and $\vec{c}$ are :

- (A) inclined at an angle of $\pi/6$ between them (B) perpendicular
(C) parallel (D) inclined at an angle of $\pi/3$ between them

यदि $(\vec{a} \times \vec{b}) \times \vec{c} = \vec{a} \times (\vec{b} \times \vec{c})$, जहाँ $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ कोई तीन ऐसे ऐसे सदिश है कि $\vec{a} \cdot \vec{b} \neq 0$, $\vec{b} \cdot \vec{c} \neq 0$, तो $\vec{a}$ तथा $\vec{c}$ है :

- (A) के बीच $\pi/6$ का कोण है (B) लम्बवत् है
(C) समान्तर है (D) के बीच $\pi/3$ का कोण है

Ans. C

Sol. $(\vec{a} \times \vec{b}) \times \vec{c} = \vec{a} \times (\vec{b} \times \vec{c})$
 $\Rightarrow (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$
 $\Rightarrow (\vec{b} \cdot \vec{c})\vec{a} = (\vec{a} \cdot \vec{b})\vec{c}$
So that $\vec{a}$ is parallel to $\vec{c}$

31. If $\vec{a} = \frac{1}{\sqrt{10}}(3\vec{i} + \vec{k})$ and $\vec{b} = \frac{1}{7}(2\vec{i} + 3\vec{j} - 6\vec{k})$, then the value of $(2\vec{a} - \vec{b}) \cdot [(\vec{a} \times \vec{b}) \times (\vec{a} + 2\vec{b})]$ is :

[AIEEE-2011]

यदि $\vec{a} = \frac{1}{\sqrt{10}}(3\vec{i} + \vec{k})$ तथा $\vec{b} = \frac{1}{7}(2\vec{i} + 3\vec{j} - 6\vec{k})$, तो $(2\vec{a} - \vec{b}) \cdot [(\vec{a} \times \vec{b}) \times (\vec{a} + 2\vec{b})]$ का मान होगा : [AIEEE-2011]

- (A) 5 (B) 3 (C) -5 (D) -3

Ans. C

Sol. $\vec{a} \cdot \vec{b} = 0$ and $|\vec{a}| = |\vec{b}| = 1$

$$\begin{aligned}
 (\vec{a} \times \vec{b}) \times (\vec{a} + 2\vec{b}) &= (\vec{a} \times \vec{b}) \times \vec{a} + (\vec{a} \times \vec{b}) \times 2\vec{b} \\
 &= -[\vec{a} \times (\vec{a} \times \vec{b}) + 2\vec{b} \times (\vec{a} \times \vec{b})] \\
 &= -[(\vec{a} \cdot \vec{b})\vec{a} - (\vec{a} \cdot \vec{a})\vec{b} + 2(\vec{b} \cdot \vec{b})\vec{a} - 2(\vec{b} \cdot \vec{a})\vec{b}] \\
 &= -[0 - \vec{b} + 2\vec{a} + 0] = [\vec{b} - 2\vec{a}] \\
 \therefore (2\vec{a} - \vec{b}) \cdot [\vec{a} \times \vec{b}) \times (\vec{a} + 2\vec{b})] \\
 &= (2\vec{a} - \vec{b}) \cdot (\vec{b} - 2\vec{a}) \\
 &= -4a^2 - b^2 + 4\vec{a} \cdot \vec{b} = -5
 \end{aligned}$$

32. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three non-zero vectors such that no two of them are collinear and $(\vec{a} \times \vec{b}) \times \vec{c} = \frac{1}{3} |\vec{b}| |\vec{c}| \vec{a}$. If

θ is the angle between vectors $\vec{b}$ and $\vec{c}$, then a value of $\sin\theta$ is : [JEE(Main)-2015]

माना $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ तीन शून्योत्तर ऐसे सदिश हैं कि उनमें से कोई दो संरेख नहीं हैं तथा $(\vec{a} \times \vec{b}) \times \vec{c} = \frac{1}{3} |\vec{b}| |\vec{c}| \vec{a}$ हैं। यदि सदिशों

$\vec{b}$ तथा $\vec{c}$ के बीच का कोण θ है, तो $\sin\theta$ का एक मान है : [JEE(Main)-2015]

(A) $\frac{2\sqrt{2}}{3}$ (B) $\frac{-\sqrt{2}}{3}$ (C) $\frac{2}{3}$ (D) $\frac{-2\sqrt{3}}{3}$

Ans. A

Sol. $(\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a} = \frac{1}{3} |\vec{b}| |\vec{c}| \vec{a}$

$$-\vec{b} \cdot \vec{c} = \frac{1}{3} (\vec{b}) \cdot (\vec{c})$$

$$\cos \theta = -\frac{1}{3}$$

$$\sin \theta = \frac{2\sqrt{2}}{3}$$

33. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three unit vectors such that $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\sqrt{3}}{2}(\vec{b} + \vec{c})$. If $\vec{b}$ is not parallel to $\vec{c}$, then the angle between $\vec{a}$ and $\vec{b}$ is : [JEE(Main)-2016]

माना $\vec{a}, \vec{b}$ तथा $\vec{c}$ तीन ऐसे मात्रक सदिश हैं कि $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\sqrt{3}}{2}(\vec{b} + \vec{c})$ है। यदि $\vec{b}, \vec{c}$ के समान्तर नहीं हैं, तो $\vec{a}$ तथा $\vec{b}$ के बीच का कोण है : [JEE(Main)-2016]

- (A) $\frac{2\pi}{3}$ (B) $\frac{5\pi}{6}$ (C) $\frac{3\pi}{4}$ (D) $\frac{\pi}{2}$

Ans. B

Sol. $(\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} = \frac{\sqrt{3}}{2}\vec{b} + \frac{\sqrt{3}}{2}\vec{c}$

$$-\vec{a} \cdot \vec{b} = \frac{\sqrt{3}}{2}$$

$$-\cos \theta = \frac{\sqrt{3}}{2}$$

$$\theta = \frac{5\pi}{6}$$

34. Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$. Let $\vec{c}$ be a vector such that $|\vec{c} - \vec{a}| = 3$, $|\vec{a} \times \vec{b}| \times \vec{c}| = 3$ and the angle between $\vec{c}$ and $\vec{a} \times \vec{b}$ be 30° . Then $\vec{a} \cdot \vec{c}$ is equal to : [JEE(Main)-2017]

माना $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ तथा $\vec{b} = \hat{i} + \hat{j}$ है। माना $\vec{c}$ एक ऐसा सदिश है कि $|\vec{c} - \vec{a}| = 3$, $|\vec{a} \times \vec{b}| \times \vec{c}| = 3$ तथा $\vec{c}$ और $\vec{a} \times \vec{b}$ के बीच कोण 30° है, तो $\vec{a} \cdot \vec{c}$ बराबर होगा : [JEE(Main)-2017]

- (A) 5 (B) $\frac{1}{8}$ (C) $\frac{25}{8}$ (D) 2

Ans. D

Sol. $\vec{a} \times \vec{b} = 2\hat{i} - 2\hat{j} + \hat{k} \Rightarrow |\vec{a} \times \vec{b}| = 3 \quad |\vec{c} - \vec{a}| = 3$

$$|\vec{a} \times \vec{b}| |\vec{c}| \sin 30^\circ = 3$$

$$c^2 + a^2 - 2\vec{a} \cdot \vec{c} = 9$$

$$|\vec{c}| = 2$$

$$\vec{c} \cdot \vec{a} = 2$$

35. A line with direction cosines proportional to 2, 1, 2 meets each of the lines $x = y + a = z$ and $x + a = 2y = 2z$. The coordinates of each of the points of intersection are given by :

एक रेखा जिसके दिक्कोज्या 2, 1, 2 के अनुपात में हैं, रेखाओं $x = y + a = z$ तथा $x + a = 2y = 2z$ को काटती हैं। इन काटे गए बिन्दुओं के निर्देशांक हैं :

- (A) (3a, 3a, 3a), (a, a, a) (B) (3a, 2a, 3a), (a, a, a)
(C) (3a, 2a, 3a), (a, a, 2a) (D) (2a, 3a, 3a), (2a, a, a)

Ans. B

Sol. Let the two lines be AB and CD having equation $\frac{x}{1} = \frac{y+a}{1} = \frac{z}{1} = \lambda$ and $\frac{x+a}{2} = \frac{y}{1} = \frac{z}{1} = \mu$

then $P \equiv (\lambda, \lambda - a, \lambda)$ and $Q \equiv (2\mu - a, \mu, \mu)$

So according to question,

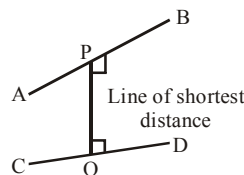
$$\frac{\lambda - 2\mu + a}{2} = \frac{\lambda - a - \mu}{1} = \frac{\lambda - \mu}{2}$$

$$\Rightarrow \mu = a \text{ and } \lambda = 3a$$

$$\therefore P \equiv (3a, 2a, 3a)$$

$$\text{and } Q \equiv (a, a, 0)$$

Trick : Put the option and check it



36. If a line makes an angle of $\pi/4$ with the positive directions of each of x-axis and y-axis, then the angle that the line makes with the positive direction of the z-axis is : [AIEEE-2007]

यदि एक रेखा प्रत्येक x- अक्ष तथा y-अक्ष की धनात्मक दिशा के साथ $\pi/4$ का कोण बनाती है, तब वह कोण, जो रेखा z-अक्ष की धनात्मक दिशा के साथ बनायेगी, होगा :

- (A) $\pi/6$ (B) $\pi/3$ (C) $\pi/4$ (D) $\pi/2$

Ans. D

Sol. $\ell = \cos \frac{\pi}{4}, m = \cos \frac{\pi}{4}$

we know that $\ell^2 + m^2 + n^2 = 1$

$$\frac{1}{2} + \frac{1}{2} + n^2 = 1 \Rightarrow n = 0$$

Hence angle with positive direction of z-axis is $\frac{\pi}{2}$

37. The angle between the lines whose direction cosines satisfy the equations $l + m + n = 0$ and $l^2 = m^2 + n^2$ is :

[JEE(Main)-2014]

दो रेखाएँ, जिनके दिक्-कोज्या, समीकरणों $l + m + n = 0$ तथा $l^2 = m^2 + n^2$ को संतुष्ट करते हैं, के बीच का कोण है :

[JEE(Main)-2014]

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{2}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{4}$

Ans. C

Sol. Solving two equation

$$l = -(m + n) \text{ then } (m + n)^2 = m^2 + n^2$$

$$2mn = 0$$

$$\text{If } m = 0, \text{ If } n = 0$$

$$l = -n \text{ or } l = -m$$

hence dir are 1, 0, -1 and 1, -1, 0

$$\cos \theta = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2}$$

$$\theta = \frac{\pi}{3}$$

38. The two lines $x = ay + b, z = cy + d$; and $x = a'y + b', z = c'y + d'$ are perpendicular to each other if :

दो रेखाएँ $x = ay + b, z = cy + d$; तथा $x = a'y + b', z = c'y + d'$ लम्बवत् होंगी, यदि और केवल यदि :

(A) $aa' + cc' = 1$

(B) $\frac{a}{a'} + \frac{c}{c'} = -1$

(C) $\frac{a}{a'} + \frac{c}{c'} = 1$

(D) $aa' + cc' = -1$

Ans. D

Sol. $\ell_1 - am_1 = 0$ and $cn_1 - n_1 = 0 \Rightarrow \frac{\ell_1}{a} = \frac{m_1}{1} = \frac{n_1}{c}$

Also $\ell_2 - a'm_2 = 0$ and $c'n_2 - n_2 = 0$

$$\Rightarrow \frac{\ell_2}{a'} = \frac{m_2}{1} = \frac{n_2}{c'}$$

$$\therefore \ell_1 \ell_2 + m_1 m_2 + n_1 n_2 = aa' + cc' + 1 = 0$$

39. If the straight lines $\frac{x-1}{k} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x-2}{3} = \frac{y-3}{k} = \frac{z-1}{2}$ intersect at a point, then the integer k is equal to : [AIEEE-2008]

यदि सरल रेखाएँ $\frac{x-1}{k} = \frac{y-2}{2} = \frac{z-3}{3}$ तथा $\frac{x-2}{3} = \frac{y-3}{k} = \frac{z-1}{2}$ एक बिन्दु पर प्रतिच्छेद करती हैं, तो पूर्णांक k का मान है:

[AIEEE-2008]

(A) 5

(B) 2

(C) -2

(D) -5

Ans. D

Sol. $\frac{x-1}{k} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x-2}{3} = \frac{y-3}{k} = \frac{z-1}{2}$

Since lines intersect in a point

$$\begin{vmatrix} k & 2 & 3 \\ 3 & k & 2 \\ 1 & 1 & -2 \end{vmatrix} = 0$$

$$-2k^2 + 5k - 25 = 0$$

$$k = -5, 5/2$$

PLANE

40. Equation of the plane containing the straight line $\frac{x}{2} = \frac{y}{3} = \frac{z}{4}$ and perpendicular to the plane containing the straight

lines $\frac{x}{3} = \frac{y}{4} = \frac{z}{2}$ and $\frac{x}{4} = \frac{y}{2} = \frac{z}{3}$, is :

रेखा T को अन्तर्विष्ट करने वाले तल का समीकरण ज्ञात कीजिये जो रेखा $\frac{x}{3} = \frac{y}{4} = \frac{z}{2}$ तथा $\frac{x}{4} = \frac{y}{2} = \frac{z}{3}$ को समाहित करने वाले तल

के लम्बवत् है :

(A) $x + 2y - 2z = 0$ (B) $3x + 2y - 2z = 0$ (C) $x - 2y + z = 0$ (D) $5x + 2y - 4z = 0$

Ans. C

Sol. Let $l_1 = \frac{x}{2} = \frac{y}{3} = \frac{z}{4}$, $l_2 = \frac{x}{3} = \frac{y}{4} = \frac{z}{2}$, $l_3 = \frac{x}{4} = \frac{y}{2} = \frac{z}{3}$

Equation of vector perpendicular to l_2 & l_3

$$NP_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & 2 \\ 4 & 2 & 3 \end{vmatrix}$$

$$= 8\hat{i} - \hat{j} - 10\hat{k}$$

Normal vector to the plane will be perpendicular to NP_2 & l_1

$$NP_1 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 8 & -1 & 10 \end{vmatrix}$$

$$= \hat{i}(-26) - \hat{j}(-52) + \hat{k}(-26)$$

$$\vec{n} = \hat{i} - 2\hat{j} + \hat{k}$$

$$\text{Plane } x - 2y + z = 0$$

41. A line with positive direction cosines passes through the point $P(2, -1, 2)$ and makes equal angles with the coordinate axes. The line meets the plane $2x + y + z = 9$ at point Q. The length of the line segment PQ equals :

एक रेखा जो बिन्दु $P(2, -1, 2)$ से गुजरती है एवं धनात्मक दिक् कोज्याएँ तथा निर्देशी अक्षों के साथ समान कोण बनाती है। रेखा तल $2x + y + z = 9$ को बिन्दु Q पर मिलती है तो रेखाखण्ड PQ की लम्बाई होगी :

- (A) 1 (B) $\sqrt{2}$ (C) $\sqrt{3}$ (D) 2

Ans. C

Sol. $(2, -1, 2) + \lambda(1, 1, 1)$
On plane $2(2 + \lambda) + (-1 + \lambda) + (2 + \lambda) = 9$
 $5 + 4\lambda = 9$
 $\lambda = 1$
Q (3, 0, 3) P(2, -1, 2)
 $PQ = \sqrt{1+1+1} = \sqrt{3}$

42. Two systems of rectangular axes have the same origin. If a plane makes intercepts a, b, c and a', b', c' on the two systems of axes respectively, then :

आयतीय अक्ष के दो निकायों के मूलबिन्दु समान हैं। यदि एक समतल अक्षों की दो निकायों पर क्रमशः a, b, c एवं a', b', c' अन्तःखण्ड काटता है, तब :

- (A) $a^2 + b^2 + c^2 = a'^2 + b'^2 + c'^2$ (B) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{a'} + \frac{1}{b'} + \frac{1}{c'}$
(C) $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{a'^2} + \frac{1}{b'^2} + \frac{1}{c'^2}$ (D) $\frac{1}{a^2 - a'^2} + \frac{1}{b^2 - b'^2} + \frac{1}{c^2 - c'^2} = 0$

Ans. C

Sol. In system I $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$
 $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$
In system II $\frac{x}{a'} + \frac{y}{b'} + \frac{z}{c'} = 1$
 $\frac{1}{a'^2} + \frac{1}{b'^2} + \frac{1}{c'^2}$

43. Let $P(3, 2, 6)$ be a point in space and Q be a point on the line $\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \mu(-3\hat{i} + \hat{j} + 5\hat{k})$. Then the value of μ for which the vector $\overrightarrow{PQ}$ is parallel to the plane $x - 4y + 3z = 1$ is :

माना $P(3, 2, 6)$ समष्टि (space) में स्थित एक बिन्दु है तथा Q रेखा $\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \mu(-3\hat{i} + \hat{j} + 5\hat{k})$ पर स्थित एक बिन्दु है।

तब μ का मान जिसके लिये सदिश $\overrightarrow{PQ}$ समतल $x - 4y + 3z = 1$ के समान्तर है, निम्न है :

- (A) $1/4$ (B) $-1/4$ (C) $1/8$ (D) $-1/8$

Ans. A

Sol. Angle between line and normal to plane is

$$\cos\left(\frac{\pi}{2} - \theta\right) = \frac{1 \times 2 - 2 \times 1 + 2\sqrt{\lambda}}{3 \times \sqrt{5 + \lambda}}, \text{ where } \theta \text{ is the angle between line and plane}$$

$$\Rightarrow \sin\theta = \frac{1 \times 2 + 2 \times (-1) + 2\sqrt{\lambda}}{3 \times \sqrt{5 + \lambda}} \Rightarrow \frac{1}{3} = \frac{2\sqrt{\lambda}}{3\sqrt{5 + \lambda}}$$

$$\Rightarrow \frac{1}{3} = \frac{2\sqrt{\lambda}}{3\sqrt{5 + \lambda}} \Rightarrow \lambda = \frac{5}{3}$$

44. If the distance of the point $P(1, -2, 1)$ from the plane $x + 2y - 2z = \alpha$, where $\alpha > 0$, is 5, then the foot of the perpendicular from P to the plane is :

यदि बिन्दु $P(1, -2, 1)$ की तल $x + 2y - 2z = \alpha$, जहाँ $\alpha > 0$ से दूरी 5 है तो बिन्दु P से तल पर डाला गया लम्ब पाद होगा :

- (A) $\left(\frac{8}{3}, \frac{4}{3}, -\frac{7}{3}\right)$ (B) $\left(\frac{4}{3}, -\frac{4}{3}, \frac{1}{3}\right)$ (C) $\left(\frac{1}{3}, \frac{2}{3}, \frac{10}{3}\right)$ (D) $\left(\frac{2}{3}, -\frac{1}{3}, \frac{5}{2}\right)$

Ans. A

Sol. $\left| \frac{1 - 4 - 2 - \alpha}{3} \right| = 5$

$$|5 + \alpha| = 15$$

$$(1, -2, 1) + \lambda(1, 2, -2)$$

$$(1 + \lambda) + 2(-2 + 2\lambda) - 2(1 - 2\lambda) = 10$$

$$1 + \lambda - 4 + 4\lambda - 2 + 4\lambda = 10$$

$$9\lambda - 5 = 0$$

$$9\lambda = 15$$

$$\lambda = \frac{5}{3}$$

$$\left(\frac{8}{3}, \frac{4}{3}, -\frac{7}{3}\right)$$

45. The distance of the point $(1, -5, 9)$ from the plane $x - y + z = 5$ measured along a straight line $x = y = z$ is :

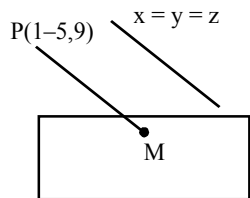
[AIEEE-2011]

बिन्दु $(1, -5, 9)$ की समतल $x - y + z = 5$ से दूरी जो सरल रेखा $x = y = z$ के अनुदिश मापी गयी है, हैं : [AIEEE-2011]

- (A) $3\sqrt{5}$ (B) $10\sqrt{3}$ (C) $5\sqrt{3}$ (D) $3\sqrt{10}$

Ans. B

Sol.



eqⁿ. of a line $\parallel$ to $x = y = z$ and passing through $(1, -5, 9)$ is

$$\frac{x-1}{1} = \frac{y+5}{1} = \frac{z-9}{1} = r$$

Let it meets plane at $M(r+1, r-5, r+9)$

Put in equation of plane

$$x - y + z = 5$$

$$r + 1 - r + 5 + r + 9 = 5$$

$$r = -10$$

Hence $M(-9, -15, -1)$

$$\text{Distance } PM = \sqrt{100 + 100 + 100} = 10\sqrt{3}$$

46. The equation of the plane containing the line $2x - 5y + z = 3$; $x + y + 4z = 5$, and parallel to the plane, $x + 3y + 6z = 1$, is : [JEE(Main)-2015]

रेखा $2x - 5y + z = 3$; $x + y + 4z = 5$ को अंतर्विष्ट करने वाले समतल, जो समतल $x + 3y + 6z = 1$ के समांतर है, का समीकरण है : [JEE(Main)-2015]

- (A) $2x + 6y + 12z = 13$ (B) $x + 3y + 6z = -7$ (C) $x + 3y + 6z = 7$ (D) $2x + 6y + 12z = -13$

Ans. C

Sol.

$$\begin{aligned} \vec{u} &= \lambda (\vec{a} \times (\vec{a} \times \vec{b})) \\ &= \lambda ((\vec{a} \cdot \vec{b})\vec{a} - (\vec{a} \cdot \vec{a})\vec{b}) \\ &= \lambda (2\vec{a} - 14\vec{b}) \\ \vec{u} &= \lambda (4\hat{i} - 8\hat{j} - 16\hat{k}) \\ \vec{u} \cdot \vec{b} &= 24 \Rightarrow \lambda = -1 \\ \Rightarrow \vec{u} &= -4\hat{i} + 8\hat{j} + 16\hat{k} \\ \Rightarrow |\vec{u}|^2 &= 16 + 64 + 256 = 336 \end{aligned}$$

47. The distance of the point $(1, 3, -7)$ from the plane passing through the point $(1, -1, -1)$ having normal perpendicular to both the lines $\frac{x-1}{1} = \frac{y+2}{-2} = \frac{z-4}{3}$ तथा $\frac{x-2}{2} = \frac{y+1}{-1} = \frac{z+7}{-1}$ is : [JEE(Main)-2017]

एक समतल जो बिन्दु $(1, -1, -1)$ से होकर जाता है तथा जिसका अभिलम्ब दोनों रेखाओं $\frac{x-1}{1} = \frac{y+2}{-2} = \frac{z-4}{3}$ तथा $\frac{x-2}{2} = \frac{y+1}{-1} = \frac{z+7}{-1}$ पर लम्ब है, की बिन्दु $(1, 3, -7)$ से दूरी है : [JEE(Main)-2017]

- (A) $\frac{5}{\sqrt{83}}$ (B) $\frac{10}{\sqrt{74}}$ (C) $\frac{20}{\sqrt{74}}$ (D) $\frac{10}{\sqrt{83}}$

Ans. D

Sol. Normal vector of plane = $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 3 \\ 2 & -1 & -1 \end{vmatrix}$

$$= 5\hat{i} + 7\hat{j} + 3\hat{k}$$

equation of plane $5x + 7y + 3z = -5$

$$\text{Distance} = \left| \frac{5(1) + 7(3) + 3(-7) + 5}{\sqrt{83}} \right|$$

$$= \frac{10}{\sqrt{83}}$$

48. If L_1 is the line of intersection of the planes $2x - 2y + 3z - 2 = 0$, $x - y + z + 1 = 0$ and L_2 is the line of intersection of the planes $x + 2y - z - 3 = 0$, $3x - y + 2z - 1 = 0$, then the distance of the origin from the plane containing the lines L_1 and L_2 is : [JEE(Main)-2018]

यदि समतलों $2x - 2y + 3z - 2 = 0$, $x - y + z + 1 = 0$ की परिच्छेदी रेखा L_1 है तथा समतलों $x + 2y - z - 3 = 0$, $3x - y + 2z - 1 = 0$ की परिच्छेदी रेखा L_2 है, तो मूल बिन्दु की दूरी उस समतल से जो रेखाओं L_1 और L_2 का अंतर्विष्ट करता है, है : [JEE(Main)-2018]

- (A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{4\sqrt{2}}$ (D) $\frac{1}{3\sqrt{2}}$

Ans. D

Sol. required plane will pass through line of intersection of planes so equation of required plane will be $x + 2y - z - 3 + \lambda (3x - y + 2z - 1) = 0$
take a fixed point on L_1 ($y = 0$)

P(-5, 0, 4)

point P will lie on required plane

$$\Rightarrow \lambda = -\frac{3}{2}$$

So equation of required plane will be

$$7x - 7y + 8z + 3 = 0$$

$$\text{distance from origin} = \left| \frac{3}{\sqrt{162}} \right| = \frac{1}{3\sqrt{2}}$$

CONCEPT OF IMAGE

49. The image of the point (-1, 3, 4) in the plane $x - 2y = 0$ is :

- (A) (15, 11, 4) (B) $\left(-\frac{17}{3}, -\frac{19}{3}, 1\right)$ (C) (8, 4, 4) (D) None of these

बिन्दु (-1, 3, 4) का तल $x - 2y = 0$ में प्रतिबिम्ब है :

- (A) (15, 11, 4) (B) $\left(-\frac{17}{3}, -\frac{19}{3}, 1\right)$ (C) (8, 4, 4) (D) इनमें से कोई नहीं

Ans. D

Sol. Equation of line PQ is

$$\frac{x+1}{1} = \frac{y-3}{-2} = \frac{z-4}{0} = \lambda$$

For some suitable value of λ , co-ordinates of point $Q(\lambda - 1, 3 - 2\lambda, 4)$

R is the mid point of P and Q.

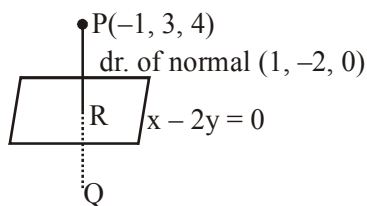
$$\therefore R \equiv \left(\frac{\lambda - 2}{2}, \frac{6 - 2\lambda}{2}, 4 \right)$$

$$R \equiv \left(\frac{\lambda}{2} - 1, 3 - \lambda, 4 \right)$$

It satisfies $x - 2y = 0$

$$\Rightarrow \lambda = \frac{14}{5}$$

$$\therefore Q = \left(\frac{2}{5}, \frac{1}{5}, 4 \right)$$



50. If the image of the point $P(1, -2, 3)$ in the plane, $2x + 3y - 4z + 22 = 0$ measured parallel to the line, $\frac{x}{1} = \frac{y}{4} = \frac{z}{5}$ is Q , then PQ is equal to : [JEE(Main)-2017]

यदि बिन्दु $P(1, -2, 3)$ का समतल $2x + 3y - 4z + 22 = 0$ में वह प्रतिबिम्ब, जो रेखा $\frac{x}{1} = \frac{y}{4} = \frac{z}{5}$ के समांतर है, Q है, तो PQ बराबर है : [JEE(Main)-2017]

- (A) $\sqrt{42}$ (B) $6\sqrt{5}$ (C) $3\sqrt{5}$ (D) $2\sqrt{42}$

Ans. D

Sol. Equation of PQ

$$\vec{r} = (1, -2, 3) + \lambda(1, 4, 5)$$

$$N(1 + \lambda, 4\lambda - 2, 3 + 5\lambda)$$

$$2(1 + \lambda) + 3(4\lambda - 2) - 4(3 + 5\lambda) + 22 = 0$$

$$6\lambda = 6$$

$$N(2, 2, 8)$$

$$\Rightarrow Q(3, 6, 13)$$

$$\Rightarrow PQ = 2\sqrt{42}$$

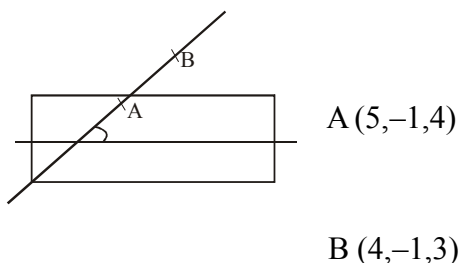
51. The length of the projection of the line segment joining the points $(5, -1, 4)$ and $(4, -1, 3)$ on the plane, $x + y + z = 7$ is : [JEE(Main)-2018]

बिन्दुओं $(5, -1, 4)$ तथा $(4, -1, 3)$ को मिलाने वाले रेखाखण्ड का समतल $x + y + z = 7$ पर डाले गए प्रक्षेप की लम्बाई है : [JEE(Main)-2018]

- (A) $\frac{1}{3}$ (B) $\sqrt{\frac{2}{3}}$ (C) $\frac{2}{\sqrt{3}}$ (D) $\frac{2}{3}$

Ans. B

Sol.



θ = angle between plane & line

$$\cos(90 - \theta) = \frac{|\vec{AB} \cdot (\hat{i} + \hat{j} + \hat{k})|}{|\vec{AB}| |\hat{i} + \hat{j} + \hat{k}|} = \frac{2}{\sqrt{6}}$$

$$\text{Req. length} = |\vec{AB}| \cos \theta$$

$$= \sqrt{2} \times \frac{\sqrt{2}}{\sqrt{6}} = \sqrt{\frac{2}{3}}$$

MISCELLANEOUS

52. Vector $\vec{x}$ satisfying the relation $\vec{A} \cdot \vec{x} = c$ and $\vec{A} \times \vec{x} = \vec{B}$ is :

सदिश $\vec{x}$ जो समीकरण $\vec{A} \cdot \vec{x} = c$ तथा $\vec{A} \times \vec{x} = \vec{B}$ को सन्तुष्ट करता है :

(A) $\frac{c\vec{A} - (\vec{A} \times \vec{B})}{|\vec{A}|}$ (B) $\frac{c\vec{A} - (\vec{A} \times \vec{B})}{|\vec{A}|^2}$ (C) $\frac{c\vec{A} + (\vec{A} \times \vec{B})}{|\vec{A}|^2}$ (D) $\frac{c\vec{A} - 2(\vec{A} \times \vec{B})}{|\vec{A}|^2}$

Ans. B

Sol. $\vec{A} \cdot \vec{x} = c$ $(\vec{A} \times \vec{x}) \times \vec{B} = 0$

$$\vec{A} \times \vec{x} = \vec{B}$$

$$(\vec{A} \times \vec{x}) \times \vec{A} = \vec{B} \times \vec{A}$$

$$\vec{x}(\vec{A} \cdot \vec{A}) - \vec{A}(\vec{A} \cdot \vec{x}) = \vec{B} \times \vec{A}$$

$$\vec{x} = \frac{c\vec{A} + \vec{B} \times \vec{A}}{|\vec{A}|^2}$$

$$\frac{c\vec{A} - (\vec{A} \times \vec{B})}{|\vec{A}|^2}$$

53. Let $\vec{a} = \hat{j} - \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} - \hat{k}$. Then the vector $\vec{b}$ satisfying $\vec{a} \times \vec{b} + \vec{c} = \vec{0}$ and $\vec{a} \cdot \vec{b} = 3$ is : [AIEEE-2010]

माना $\vec{a} = \hat{j} - \hat{k}$ तथा $\vec{c} = \hat{i} - \hat{j} - \hat{k}$ तो सदिश $\vec{b}$ जो $\vec{a} \times \vec{b} + \vec{c} = \vec{0}$ तथा $\vec{a} \cdot \vec{b} = 3$ को संतुष्ट करता है : [AIEEE-2010]

(A) $-\hat{i} + \hat{j} - 2\hat{k}$ (B) $2\hat{i} - \hat{j} + 2\hat{k}$ (C) $\hat{i} - \hat{j} - 2\hat{k}$ (D) $\hat{i} + \hat{j} - 2\hat{k}$

Ans. A

Sol. $(\vec{a} \times \vec{b}) + \vec{c} = \vec{0}$

$$(\vec{a} \times \vec{b}) = -\vec{c}$$

$$\Rightarrow \vec{a} \times (\vec{a} \times \vec{b}) = -\vec{a} \times \vec{c}$$

$$\Rightarrow (\vec{a} \cdot \vec{b})\vec{a} - |\vec{a}|^2 \vec{b} = -\vec{a} \times \vec{c}$$

$$\Rightarrow 3(\vec{j} - \vec{k}) - 2\vec{b} = -(-2\vec{i} - \vec{j} - \vec{k}) \quad (\vec{a} \times \vec{c} = -2\vec{i} - \vec{j} - \vec{k})$$

$$\Rightarrow 2\vec{b} = (-2\vec{i} + 2\vec{j} - 4\vec{k})$$

$$\Rightarrow \vec{b} = -\vec{i} + \vec{j} - 2\vec{k}$$

54. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are linearly independent vectors, then which one of the following set of vectors is linearly dependent?

(A) $\vec{a} + \vec{b}$, $\vec{b} + \vec{c}$, $\vec{c} + \vec{a}$ (B) $\vec{a} - \vec{b}$, $\vec{b} - \vec{c}$, $\vec{c} - \vec{a}$ (C) $\vec{a} \times \vec{b}$, $\vec{b} \times \vec{c}$, $\vec{c} \times \vec{a}$ (D) None

यदि $\vec{a}$, $\vec{b}$, $\vec{c}$ रेखीय रूप से स्वतंत्र सदिश हैं, तो निम्न में से सदिशों का कौनसा समुच्चय रेखीय रूप से परतंत्र है?

(A) $\vec{a} + \vec{b}$, $\vec{b} + \vec{c}$, $\vec{c} + \vec{a}$ (B) $\vec{a} - \vec{b}$, $\vec{b} - \vec{c}$, $\vec{c} - \vec{a}$ (C) $\vec{a} \times \vec{b}$, $\vec{b} \times \vec{c}$, $\vec{c} \times \vec{a}$ (D) कोई नहीं

Ans. (B)

Sol. $(\vec{a} \times \vec{b}) + \vec{c} = 0$

$$(\vec{a} \times \vec{b}) = -\vec{c}$$

$$\Rightarrow \vec{a} \times (\vec{a} \times \vec{b}) = -\vec{a} \times \vec{c}$$

$$\Rightarrow (\vec{a} \cdot \vec{b})\vec{a} - |\vec{a}|^2 \vec{b} = -\vec{a} \times \vec{c}$$

$$\Rightarrow 3(\vec{j} - \vec{k}) - 2\vec{b} = -(-2\vec{i} - \vec{j} - \vec{k}) \quad (\vec{a} \times \vec{c} = -2\vec{i} - \vec{j} - \vec{k})$$

$$\Rightarrow 2\vec{b} = (-2\vec{i} + 2\vec{j} - 4\vec{k})$$

$$\Rightarrow \vec{b} = -\vec{i} + \vec{j} - 2\vec{k}$$

Exercise J-2

BASIC PROPERTIES OF VECTORS

1. If $\vec{a}$ and $\vec{b}$ are non collinear vectors such that, $\vec{p} = (x + 4y)\vec{a} + (2x + y + 1)\vec{b}$ and $\vec{q} = (y - 2x + 2)\vec{a} + (2x - 3y - 1)\vec{b}$, find x and y such that $3\vec{p} = 2\vec{q}$.

यदि $\vec{a}$ तथा $\vec{b}$ असंरेखीय सदिश इस प्रकार हैं कि $\vec{p} = (x + 4y)\vec{a} + (2x + y + 1)\vec{b}$ तथा $\vec{q} = (y - 2x + 2)\vec{a} + (2x - 3y - 1)\vec{b}$, यदि $3\vec{p} = 2\vec{q}$ है, तो x एवं y का मान ज्ञात कीजिए।

Ans. $x = 2, y = -1$

Sol. $\vec{P} = (x + 4y)\vec{a} + (2x + y + 1)\vec{b}$
 $\vec{q} = (y - 2x + 2)\vec{a} + (2x - 3y - 1)\vec{b}$
 $3\vec{P} = 2\vec{q}$
 $3(x + 4y) = 2(y - 2x + 2)$
 $7x + 10y = 4$
 $6x + 3y + 3 = 4x - 6y - 2$
 $2x + 9y = -5$
 $5x + y = 9$

$$\left. \begin{array}{l} 7x + 10y = 4 \\ 6x + 3y + 3 = 4x - 6y - 2 \\ 2x + 9y = -5 \\ 5x + y = 9 \end{array} \right\} \begin{array}{l} x = 2 \\ y = -1 \end{array}$$

2. Show that the points $\vec{a} - 2\vec{b} + 3\vec{c}$; $2\vec{a} + 3\vec{b} - 4\vec{c}$ and $-7\vec{b} + 10\vec{c}$ are collinear.

सिद्ध कीजिए कि बिन्दु $\vec{a} - 2\vec{b} + 3\vec{c}$; $2\vec{a} + 3\vec{b} - 4\vec{c}$ तथा $-7\vec{b} + 10\vec{c}$ संरेखीय हैं।

Ans.

Sol. $\left. \begin{array}{l} \vec{p} = \vec{a} - 2\vec{b} + 3\vec{c} \\ \vec{q} = 2\vec{a} + 3\vec{b} - 4\vec{c} \\ \vec{r} = -7\vec{b} + 10\vec{c} \end{array} \right\} \text{Collinear}$
 $2\vec{p} - \vec{q} = \vec{r}$
 $\vec{qp} = -\vec{a} - 5\vec{b} + 7\vec{c}$
 $\vec{rq} = 2\vec{a} + 10\vec{b} + 14\vec{c}$
 $\vec{qp} = \lambda \vec{rq}$

3. Points X and Y are taken on the sides QR and RS, respectively of a parallelogram PQRS, so that $\vec{QX} = 4\vec{XR}$ and $\vec{RY} = 4\vec{YS}$. The line XY cuts the line PR at Z. Prove that $\vec{PZ} = \left(\frac{21}{25}\right)\vec{PR}$.

समान्तर चतुर्भुज PQRS की भुजा QR तथा RS पर बिन्दु X एवं Y इस प्रकार से है कि $\vec{QX} = 4\vec{XR}$ तथा $\vec{RY} = 4\vec{YS}$. यदि रेखा

XY रेखा PR को Z पर काटती है तो सिद्ध करो कि $\vec{PZ} = \left(\frac{21}{25}\right)\vec{PR}$.

Ans.

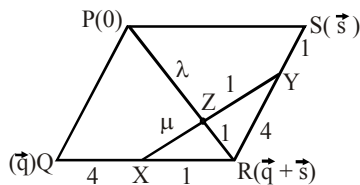
Sol. $\vec{QX} = 4\vec{XR}$

$\vec{RY} = 4\vec{YS}$

Let $\vec{p}$ be origin

and $R(\vec{q} + \vec{s})$

from figure



P.V. of X = $\frac{4(\vec{q} + \vec{s}) + \vec{q}}{5} = \frac{5\vec{q} + 4\vec{s}}{5}$

P.V. of Y = $\frac{4\vec{s} + \vec{q} + \vec{s}}{5} = \frac{5\vec{s} + \vec{q}}{5}$

Now Let Z divides PR in ratio $\lambda : 1$

Now Let Z divides XY in ratio $\mu : 1$

P.V. of Z = $\frac{\lambda(\vec{q} + \vec{s})}{\lambda + 1}$ (from PR)

P.V. of Z = $\frac{\mu(5\vec{s} + \vec{q})}{\mu + 1} + \frac{5\vec{q} + 4\vec{s}}{5}$ (from XY)

equating both Z then we get

$\frac{\lambda}{\lambda + 1} = \frac{\mu + 5}{5(\mu + 1)}$ (i)

$\frac{\lambda}{\lambda + 1} = \frac{5\mu + 4}{5(\mu + 1)}$ (ii)

from (i) & (ii), $\mu = \frac{1}{4}$ and $\lambda = \frac{21}{4}$

So P.V. of Z = $\frac{\frac{21}{4}}{\frac{21}{4} + 1} (\vec{q} + \vec{s})$

$= \frac{21}{25} (\vec{q} + \vec{s}) = \frac{21}{25} \vec{PR}$

4. In a $\triangle ABC$, points E and F divide sides AC and AB respectively so that $\frac{AE}{EC} = 4$ and $\frac{AF}{FB} = 1$. Suppose D is a point on side BC. Let G be the intersection of EF and AD and suppose D is situated so that $\frac{AG}{GD} = \frac{3}{2}$. If the ratio $\frac{BD}{DC} = \frac{a}{b}$, where a and b are in their lowest form, find the value of (a + b).

त्रिभुज $\triangle ABC$ में बिन्दु E एवं F ऐसा AC एवं AB को इस प्रकार से विभाजित करती है कि $\frac{AE}{EC} = 4$ तथा $\frac{AF}{FB} = 1$ है। रेखा BC

पर D एक बिन्दु है। यदि EF तथा AD का प्रतिच्छेद बिन्दु G है तथा D इस प्रकार से है कि $\frac{AG}{GD} = \frac{3}{2}$ यदि $\frac{BD}{DC} = \frac{a}{b}$ जहाँ a एवं b का महत्तम समापवर्तक 1 है तो (a + b) का मान बताओं।

Ans. 9

Sol. $\frac{BD}{DC} = \frac{a}{b} \Rightarrow a + b$

$$F\left(\frac{\vec{b}}{2}\right) \quad E\left(\frac{4\vec{c}}{5}\right) \quad G\left(\frac{3\vec{d}}{5}\right) \quad \vec{d} = \frac{\alpha\vec{c} + \vec{b}}{\alpha + 1}$$

$$\vec{G} = \frac{\left(\frac{4\mu\vec{c}}{5} + \frac{\vec{b}}{2}\right)}{\mu + 1} = \frac{3}{5} \left(\frac{\alpha\vec{c} + \vec{b}}{\alpha + 1}\right)$$

$$\frac{4\mu}{5(\mu + 1)} = \frac{3\alpha}{5(\alpha + 1)}$$

$$4\mu\alpha + 4\mu = 3\mu\alpha + 3\alpha$$

$$\mu\alpha = 3\alpha - 4\mu$$

$$\frac{1}{2(\mu + 1)} = \frac{3}{5(\alpha + 1)}$$

$$5\alpha + 5 = 6\mu + 6$$

$$5\alpha - 6\mu = 1$$

$$5\alpha = 1 + 6\mu$$

$$2\mu(5\alpha) = 3(5\alpha) - 20\mu$$

$$2\mu + 12\mu^2 = 3 + 18\mu - 20\mu$$

$$12\mu^2 + 4\mu - 3 = 0$$

$$\mu = \frac{-4 \pm \sqrt{16 + 12 \times 12}}{2 \times 12}$$

$$6\mu\alpha = 18\alpha - 4 \cdot 6\mu$$

$$5\alpha^2 - \alpha = 18\alpha - 20\alpha + 4$$

$$5\alpha^2 - \alpha - 4 = 0$$

$$5\alpha^2 - 5\alpha + 4\alpha - 4 = 0$$

$$\alpha = 1$$

$$\alpha = -\frac{4}{9}$$

DOT PRODUCT AND ITS APPLICATION

5. Find a unit vector $\hat{a}$ which makes an angle $(\pi/4)$ with axis of z and is such that $\hat{a} + \hat{i} + \hat{j}$ is a unit vector.

एक इकाई सदिश $\hat{a}$ ऐसा ज्ञात करो जो कि z अक्ष से $(\pi/4)$ का कोण बनाता है तथा $\hat{a} + \hat{i} + \hat{j}$ एक इकाई सदिश है।

Ans. $\frac{-1}{2}\hat{i} - \frac{1}{2}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$

Sol. Let $\hat{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

$$\hat{a} \cdot \hat{k} = \cos \frac{\pi}{4}$$

$$a_3 = \frac{1}{\sqrt{2}}$$

$$a_1^2 + a_2^2 = \frac{1}{2}$$

$$\hat{a} + \hat{i} + \hat{j} = (a_1 + 1)\hat{i} + (a_2 + 1)\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$$

$$(a_1 + 1)^2 + (a_2 + 1)^2 = \frac{1}{2}$$

$$2a_1 + 1 + 2a_2 + 1 = 0$$

$$a_1 + a_2 = -1 \quad \dots(1)$$

$$2a_1a_2 = +\frac{1}{2}$$

$$a_1a_2 = \frac{1}{4} \quad \dots(2)$$

from (1) & (2)

$$a_1 = -\frac{1}{2}$$

$$\text{and } a_2 = -\frac{1}{2}$$

6. Prove that $\left(\frac{\vec{a}}{a^2} - \frac{\vec{b}}{b^2} \right)^2 = \left(\frac{\vec{a} - \vec{b}}{|\vec{a}| |\vec{b}|} \right)^2$

सिद्ध करो कि $\left(\frac{\vec{a}}{a^2} - \frac{\vec{b}}{b^2} \right)^2 = \left(\frac{\vec{a} - \vec{b}}{|\vec{a}| |\vec{b}|} \right)^2$

Ans.

Sol.

$$\left(\frac{\vec{a}}{(\vec{a})^2} - \frac{\vec{b}}{(\vec{b})^2} \right) = \left(\frac{\vec{a}|\vec{b}| - \vec{b}|\vec{a}|}{|\vec{a}|^2 \cdot |\vec{b}|^2} \right)^2$$

$$= \left(\frac{\vec{a} \cdot \vec{b} (\vec{b} - \vec{a})}{(\vec{a} \cdot \vec{b})^2} \right)^2$$

$$= \frac{\vec{b} - \vec{a}}{\vec{a}}$$

7. If $\vec{a}$ and $\vec{b}$ are any two unit vectors, then find the range of $\frac{3|\vec{a} + \vec{b}|}{2} + 2|\vec{a} - \vec{b}|$.

यदि $\vec{a}$ एवं $\vec{b}$ दो सदिश है तो $\frac{3|\vec{a} + \vec{b}|}{2} + 2|\vec{a} - \vec{b}|$ का परिसर बताओं।

Ans. Range: [3, 5]

Sol.

$$\frac{3}{2}|\vec{a} + \vec{b}| + 2|\vec{a} - \vec{b}| \quad |\vec{a} + \vec{b}| = 2 \cos \frac{\alpha}{2}$$

$$|\vec{a} - \vec{b}| = 2 \sin \frac{\alpha}{2}$$

$$3 \cos \frac{\alpha}{2} + 2 \sin \frac{\alpha}{2}$$

$$[3, 5]$$

8. Let $\vec{V}_1$ and $\vec{V}_2$ are two vectors such that $\vec{V}_1 = 2(\sin \alpha + \cos \alpha) \hat{i} + \hat{j}$ and $\vec{V}_2 = \sin \beta \hat{i} + \cos \beta \hat{j}$ where α and β satisfy the relation $2(\sin \alpha + \cos \alpha) \sin \beta = 3 - \cos \beta$, find the value of $(3 \tan^2 \alpha + 4 \tan^2 \beta)$.

माना $\vec{V}_1$ तथा $\vec{V}_2$ दो सदिश इस प्रकार है कि $\vec{V}_1 = 2(\sin \alpha + \cos \alpha) \hat{i} + \hat{j}$ तथा $\vec{V}_2 = \sin \beta \hat{i} + \cos \beta \hat{j}$ जहाँ α तथा β समीकरण $2(\sin \alpha + \cos \alpha) \sin \beta = 3 - \cos \beta$, को सन्तुष्ट करते है तो $(3 \tan^2 \alpha + 4 \tan^2 \beta)$ का मान होगा :

Ans. 35

Sol. We have $\vec{V}_1 \cdot \vec{V}_2 = 2(\sin \alpha + \cos \alpha) \sin \beta + \cos \beta$

$$\therefore \vec{V}_1 \cdot \vec{V}_2 = 3 \quad [\text{given that } 2(\sin \alpha + \cos \alpha) \sin \beta + \cos \beta = 3]$$

$$|\vec{V}_1|^2 |\vec{V}_2|^2 \cos^2 \theta = 9 \quad (\theta = \vec{V}_1 \wedge \vec{V}_2)$$

$$[4(1 + \sin 2\alpha) + 1] (1)\cos^2\theta = 9$$

$$(5 + 4 \sin 2\alpha) \cos^2\theta = 9$$

$$\cos^2\theta = \frac{9}{5 + 4 \sin 2\alpha} \leq 1 \Rightarrow 9 \leq 5 + 4 \sin 2\alpha \Rightarrow 4 \leq 4 \sin 2\alpha$$

$$\therefore \sin 2\alpha = 1 \Rightarrow \alpha = \frac{\pi}{4}$$

$$\therefore \cos^2\theta = \frac{9}{9} = 1 \Rightarrow \theta = 0$$

$$\text{Hence } \theta = 0 \Rightarrow \vec{V}_1 \text{ and } \vec{V}_2 \text{ are collinear i.e. } \frac{2(\sin \alpha + \cos \alpha)}{\sin \beta} = \frac{1}{\cos \beta} \quad \left(\alpha = \frac{\pi}{4} \right)$$

$$\tan \beta = 2 \left(\frac{2}{\sqrt{2}} \right) = 2\sqrt{2}$$

$$\text{Hence } 3 \tan^2\alpha + 4 \tan^2\beta = 3 + (4)(8) = 35 \text{ Ans.}$$

9. In a unit cube. Find :

- The angle between the diagonal of the cube and a diagonal of a face skew to it.
- The angle between the diagonals of two faces of the cube through the same vertex.
- The angle between a diagonal of a cube and a diagonal of a face intersecting it.

एक इकाई घन में ज्ञात कीजिए :

- एक काय विकर्ण के तथा फलक विकर्ण, जो इसके साथ विषममतीय (skew) है, के मध्य कोण बताओं ।
- एक ही शीर्ष से जाने वाले दो समतल (face) के विकर्ण के बीच कोण बताओ ।
- घन के काय विकर्ण तथा फलक विकर्ण, जो इसको प्रतिच्छेद करता है, के मध्य कोण बताओ ।

Ans. (a) $\cot^{-1}(0)$ (b) $\cot^{-1} \frac{1}{\sqrt{3}}$ (c) $\cot^{-1} \sqrt{2}$

Sol. $\vec{BE} = (\vec{a}, \vec{a}, -\vec{a})$

$$\vec{GD} = (\vec{a}, -\vec{a}, \vec{a})$$

$$\vec{BE} \cdot \vec{GD} = a^2 - a^2 - a^2$$

$$\sqrt{3}a \cdot \sqrt{3}a \cdot \cos \theta = a^2$$

$$\cos \theta = \frac{1}{3}$$

10. Given that $\vec{u} = \hat{i} - 2\hat{j} + 3\hat{k}$; $\vec{v} = 2\hat{i} + \hat{j} + 4\hat{k}$; $\vec{w} = \hat{i} + 3\hat{j} + 3\hat{k}$ and $(\vec{u} \cdot \vec{R} - 10)\hat{i} + (\vec{v} \cdot \vec{R} - 20)\hat{j} + (\vec{w} \cdot \vec{R} - 20)\hat{k} = \vec{0}$.

Find the unknown vector $\vec{R}$.

दिया है कि $\vec{u} = \hat{i} - 2\hat{j} + 3\hat{k}$; $\vec{v} = 2\hat{i} + \hat{j} + 4\hat{k}$; $\vec{w} = \hat{i} + 3\hat{j} + 3\hat{k}$ तथा $(\vec{u} \cdot \vec{R} - 10)\hat{i} + (\vec{v} \cdot \vec{R} - 20)\hat{j} + (\vec{w} \cdot \vec{R} - 20)\hat{k} = 0$ तो सदिश $\vec{R}$ ज्ञात कीजिए।

Ans. $-\hat{i} + 2\hat{j} + 5\hat{k}$

Sol. $\vec{u}(1, -2, 3)$ $\vec{R}(\alpha, \beta, \gamma)$ $\vec{v}(2, 1, 4)$ $\vec{w}(1, 3, 3)$

$$\vec{u} \cdot \vec{R} = 10, \vec{v} \cdot \vec{R} = 20, \vec{w} \cdot \vec{R} = 20$$

$$\alpha - 2\beta + 3\gamma = 10$$

$$2\alpha + \beta + 4\gamma = 20$$

$$\alpha + 3\beta + 3\gamma - 20$$

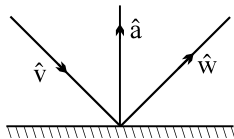
$$\alpha = -1$$

$$\beta = 2$$

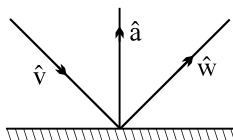
$\gamma = 5$

LINEAR COMBINATION THEOREM IN A PLANE

11. Incident ray is along the unit vector $\hat{v}$ and the reflected ray is along the unit vector $\hat{w}$. The normal is along unit vector $\hat{a}$ outwards. Express $\hat{w}$ in terms of $\hat{a}$ and $\hat{v}$.

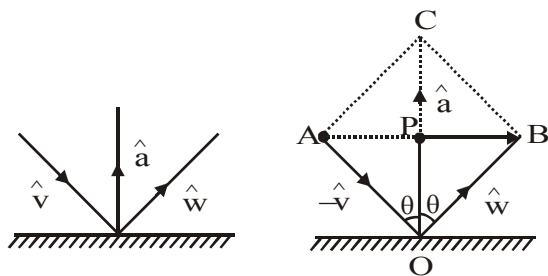


आवर्तित किरण इकाई सदिश $\hat{v}$ के अनुदिश तथा परावर्तित किरण इकाई सदिश $\hat{w}$ के अनुदिश है। अभिलम्ब इकाई सदिश $\hat{a}$ के अनुदिश बाहर की तरफ है। $\hat{w}$ को $\hat{a}$ और $\hat{v}$ के पदों में व्यक्त कीजिये।



Ans. $\hat{w} = \hat{v} - 2(\hat{a} \cdot \hat{v}) \hat{a}$

Sol. Given that incident ray is along $\hat{v}$, reflected ray is along $\hat{w}$ and normal is along $\hat{a}$, outwards. The given figure can be redrawn as shown.



We know that incident ray, reflected ray and normal lie in a plane, and angle of incidence = angle of reflection.

Therefore $\hat{a}$ will be along the angle bisector of $\hat{w}$ and $-\hat{v}$, i.e.,

$$\hat{a} = \frac{\hat{w} + (-\hat{v})}{|\hat{w} - \hat{v}|} \quad \dots(i)$$

[$\because$ angle bisector will along a vector dividing in same ratio as the ratio of the sides forming that angle.]

But $\hat{a}$ is a unit vector

where $|\hat{w} - \hat{v}| = OC = 2OP$

$$= 2 |\hat{w}| \cos \theta = 2 \cos \theta$$

Substituting this value in equation (i) we get

$$\hat{a} = \frac{\hat{w} - \hat{v}}{2 \cos \theta}$$

$$\begin{aligned} \therefore \hat{w} &= \hat{v} + (2 \cos \theta) \hat{a} \\ &= \hat{v} - 2(\hat{a} \cdot \hat{v}) \hat{a} \quad [\because \hat{a} \cdot \hat{v} = -\cos \theta]. \end{aligned}$$

CROSS PRODUCT AND ITS APPLICATION

12. Find the minimum area of the triangle whose vertices are $A(-1, 1, 2)$; $B(1, 2, 3)$ and $C(t, 1, 1)$ where t is a real number.

यदि किसी त्रिभुज ABC का शीर्ष यदि $A(-1, 1, 2)$; $B(1, 2, 3)$ एवं $C(t, 1, 1)$ है जहाँ t एक वास्तविक संख्या है तो त्रिभुज का न्यूनतम क्षेत्रफल बताओं।

Ans. $\frac{\sqrt{3}}{2}$

Sol.

$$\begin{aligned} \text{Ar} &= \frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ t-1 & -1 & -2 \\ -2 & -1 & -1 \end{vmatrix} \\ &= \frac{1}{2} \hat{i}(-1) - \hat{j}(1-t-4) + \hat{k}(1-t-2) \\ &= \frac{1}{2}(-1, +t+3, -t-1) \\ \text{Ar} &= \frac{1}{2} \sqrt{1+(t+3)^2+(t+1)^2} = \frac{1}{2} \sqrt{2t^2+8t+11} \\ 4\text{Ar}^2 &= 2t^2+8t+11 \\ \frac{d(4\text{Ar}^2)}{dx} &= 4t+8 \end{aligned}$$

$$4t + 8 = 0$$

$$\Rightarrow t = -2$$

$$Ar = \frac{1}{2} \sqrt{2t^2 + 8t + 11}$$

$$Ar = \frac{1}{2} \sqrt{3}$$

13. Let $\overrightarrow{OA} = \vec{a}$; $\overrightarrow{OB} = 100\vec{a} + 2\vec{b}$ and $\overrightarrow{OC} = \vec{b}$ where O, A, and C are non collinear points. Let P denotes the area of the parallelogram with $\overrightarrow{OA}$ and $\overrightarrow{OC}$ as adjacent sides and Q denotes the area of the quadrilateral OABC. If $Q = \lambda P$. Find the value of λ .

माना $\overrightarrow{OA} = \vec{a}$; $\overrightarrow{OB} = 100\vec{a} + 2\vec{b}$ तथा $\overrightarrow{OC} = \vec{b}$ जहाँ O, A, तथा C विषम रेखीय बिन्दु है। माना P एक समान्तर चतुर्भुज, जिसकी आसन्न भुजाएँ $\overrightarrow{OA}$ तथा $\overrightarrow{OC}$ हैं, का क्षेत्रफल है तथा Q चतुर्भुज OABC का क्षेत्रफल है यदि $Q = \lambda P$ तो λ का मान ज्ञात कीजिये।

Ans. 51

Sol. $P = |(\overrightarrow{OA}) \times (\overrightarrow{OC})|$

$$P = |\vec{a} \times \vec{b}|$$

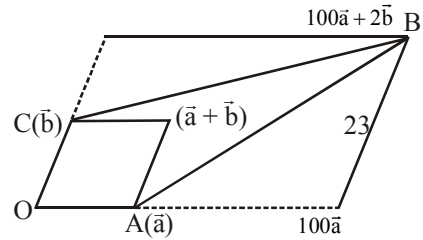
$$Q = \frac{1}{2} |\overrightarrow{OB} \times \overrightarrow{AC}| = \frac{1}{2} |(100\vec{a} + 2\vec{b}) \times (\vec{a} - \vec{b})|$$

$$= \frac{1}{2} |-100\vec{a} \times \vec{b} + 2\vec{b} \times \vec{a}| = \frac{1}{2} |102(\vec{a} \times \vec{b})| = 51 |\vec{a} \times \vec{b}|$$

Now $Q = \lambda P$

$$51 |\vec{a} \times \vec{b}| = \lambda |\vec{a} \times \vec{b}|$$

$\lambda = 51$ Ans.



[FOT]

14. Let O be the interior point of ΔABC such that $2\overrightarrow{OA} + 3\overrightarrow{OB} + 6\overrightarrow{OC} = 0$. If $\frac{\text{Ar. of } \Delta ABC}{\text{Ar. of } \Delta AOB} = \frac{m}{n}$ where m and n are relatively prime, then find $(m + n)$.

माना O ΔABC में स्थित एक आन्तरिक बिन्दु है इस प्रकार कि $2\overrightarrow{OA} + 3\overrightarrow{OB} + 6\overrightarrow{OC} = 0$ यदि $\frac{\text{Ar. of } \Delta ABC}{\text{Ar. of } \Delta AOB} = \frac{m}{n}$ जहाँ m

तथा n सापेक्षिक अभाज्य है तो $(m + n)$ का मान ज्ञात कीजिये।

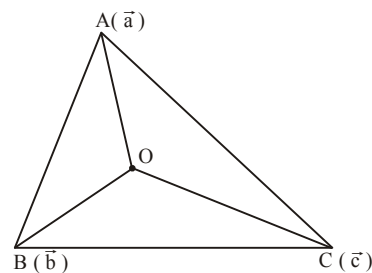
Ans. 17

Sol. Area of $\Delta ABC = \frac{1}{2} |\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|$

Now given $2\vec{a} + 3\vec{b} + 6\vec{c} = 0$

Cross with $\vec{a}$, $3\vec{a} \times \vec{b} + 6\vec{a} \times \vec{c} = 0$
 or $\vec{a} \times \vec{b} = 2(\vec{c} \times \vec{a}) = 0$ (1)

again cross with $\vec{b}$, $2\vec{a} \times \vec{b} + 6\vec{c} \times \vec{b} = 0$
 or $\vec{a} \times \vec{b} = 3(\vec{b} \times \vec{c}) = 0$ (2)



Area of $\Delta OAB = \frac{1}{2} |\vec{a} \times \vec{b}|$ [FOT]

$$\frac{\text{Ar. of } \Delta ABC}{\text{Ar. of } \Delta AOB} = \frac{\frac{1}{2} |\vec{a} \times \vec{b}| \left(1 + \frac{1}{2} + \frac{1}{3}\right)}{\frac{1}{2} |\vec{a} \times \vec{b}|} = \frac{11}{6} \Rightarrow m + n = 17 \text{ Ans.}$$

15. Determine vector of magnitude 9 which is perpendicular to both the vectors : $4\hat{i} - \hat{j} + 3\hat{k}$ and $-2\hat{i} + \hat{j} - 2\hat{k}$.

एक ऐसे सदिश जो बताओं जिसका परिमाण 9 है एवं वह दोनों सदिश $4\hat{i} - \hat{j} + 3\hat{k}$ तथा $-2\hat{i} + \hat{j} - 2\hat{k}$ के लम्बवत् है।

Ans. $\pm 3(\hat{i} - 2\hat{j} - 2\hat{k})$

Sol.
$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -1 & 3 \\ -2 & 1 & -2 \end{vmatrix} = \hat{i}(-1) - \hat{j}(-2) + \hat{k}(+2)$$

$$= (-1, +2, 2)$$

$$\vec{a} = \pm 3(-1, 2, 2)$$

16. A triangle has vertices $(1, 1, 1)$; $(2, 2, 2)$, $(1, 1, y)$ and has the area equal to $\operatorname{cosec} \left(\frac{\pi}{4} \right)$ sq. units. Find the value of y.

एक त्रिभुज का शीर्ष $(1, 1, 1)$; $(2, 2, 2)$ एवं $(1, 1, y)$ है तथा उसका क्षेत्रफल $\operatorname{cosec} \left(\frac{\pi}{4} \right)$ वर्ग मात्रक है तो y का मान बताओं।

Ans. $y = 3$ or $y = -1$

Sol. $\text{Arg} = \sqrt{2}$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & 1 & 2y \end{vmatrix} = 2\sqrt{2}$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 0 & 0 & 1-y \end{vmatrix}$$

$$|(1-y)(\hat{i}-\hat{j})| = 2\sqrt{2}$$

$$(1-y)\sqrt{2} = \pm 2\sqrt{2}$$

$$y = 1 \pm 2$$

$$= 3, -1$$

17. Given that $\vec{a}$ and $\vec{b}$ are two unit vectors such that angle between $\vec{a}$ and $\vec{b}$ is $\cos^{-1}\left(\frac{1}{4}\right)$. If $\vec{c}$ be a vector in the

plane of $\vec{a}$ and $\vec{b}$, such that $|\vec{c}| = 4$, $\vec{c} \times \vec{b} = 2\vec{a} \times \vec{b}$ and $\vec{c} = \lambda\vec{a} + \mu\vec{b}$ then, find :

- (a) the value of λ (b) the sum of values of μ and (c) the product of all possible values of μ

$\vec{a}$ एवं $\vec{b}$ इकाई सदिश है तथा इनके बीच का कोण $\cos^{-1}\left(\frac{1}{4}\right)$ है यदि उस तल में एक सदिश $\vec{c}$ इस प्रकार से है कि

$|\vec{c}| = 4$, $\vec{c} \times \vec{b} = 2\vec{a} \times \vec{b}$ तथा $\vec{c} = \lambda\vec{a} + \mu\vec{b}$ तो :

- (a) λ का मान बताओं (b) μ के सभी मानों का योग बताओं (c) μ के सारे मानों का गुणनफल बताओं

Ans. (a) 2 (b) -1 (c) -12

Sol. $\vec{c} \times \vec{b} = 2\vec{a} \times \vec{b}$

$$(\vec{c} - 2\vec{a}) \times \vec{b} = 0$$

$$\vec{c} = 2\vec{a} + \mu\vec{b}$$

$$\lambda = 2$$

$$|\vec{c}| = 4$$

$$c^2 = 4a^2 + \mu^2 b^2 + 4\mu \times \frac{1}{4}$$

$$\mu^2 + 4 + \mu = 16$$

$$\mu^2 + \mu - 12 = 0$$

$$\mu = -4, 3$$

18. Let $\vec{a}$, $\vec{b}$, $\vec{c}$, $\vec{d}$ are four distinct vectors satisfying $\vec{a} \times \vec{b} = \vec{c} \times \vec{d}$ and $\vec{a} \times \vec{c} = \vec{b} \times \vec{d}$. Show that $\vec{a} \cdot \vec{b} + \vec{c} \cdot \vec{d} \neq \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{d}$.

माना $\vec{a}$, $\vec{b}$, $\vec{c}$, $\vec{d}$ चार भिन्न सदिश हैं जो $\vec{a} \times \vec{b} = \vec{c} \times \vec{d}$ तथा $\vec{a} \times \vec{c} = \vec{b} \times \vec{d}$ को सन्तुष्ट करते हैं। सिद्ध कीजिए

$$\vec{a} \cdot \vec{b} + \vec{c} \cdot \vec{d} \neq \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{d}$$

Ans.

Sol. Given that $\vec{a} \neq \vec{b} \neq \vec{c} \neq \vec{d}$

Such that $\vec{a} \times \vec{c} = \vec{b} \times \vec{d} \quad \dots(i)$

$$\vec{a} \times \vec{b} = \vec{c} \times \vec{d} \quad \dots(ii)$$

To prove that $(\vec{a} - \vec{d}) \cdot (\vec{b} - \vec{c}) \neq 0$

Subtracting equation (ii) from (i) we get

$$\vec{a} \times (\vec{c} - \vec{b}) = (\vec{b} - \vec{c}) \times \vec{d}$$

$$\Rightarrow \vec{a} \times (\vec{c} - \vec{b}) = \vec{d} \times (\vec{c} - \vec{b})$$

$$\Rightarrow \vec{a} \times (\vec{c} - \vec{b}) - \vec{d} \times (\vec{c} - \vec{b}) = 0$$

$$\Rightarrow (\vec{a} - \vec{d}) \times (\vec{c} - \vec{b}) = 0 \Rightarrow \vec{a} - \vec{d} \parallel \vec{c} - \vec{b}$$

[$\because \vec{a} - \vec{d} \neq 0, \vec{c} - \vec{b} \neq 0$ as all distinct]

$\Rightarrow$ Angle between $\vec{a} - \vec{d}$ and $\vec{c} - \vec{b}$ is either 0 or 180° .

$$\Rightarrow (\vec{a} - \vec{d}) \cdot (\vec{c} - \vec{b}) = |\vec{a} - \vec{d}| |\vec{c} - \vec{b}|$$

[$\cos 0^\circ$ or $\cos 180^\circ$] $\neq 0$ as $\vec{a}, \vec{d}, \vec{c}, \vec{b}$ all are different.

19. Let $\vec{a} = -\hat{i} - \hat{k}, \vec{b} = -\hat{i} + \hat{j}$ and $\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} \cdot \vec{a} = 0$, then the value of $\vec{r} \cdot \vec{b}$ is : [JEE-2011]

माना $\vec{a} = -\hat{i} - \hat{k}, \vec{b} = -\hat{i} + \hat{j}$ तथा $\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$ तीन सदिश है यदि सदिश $\vec{r}$ इस प्रकार है कि $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ तथा $\vec{r} \cdot \vec{a} = 0$ है, तो $\vec{r} \cdot \vec{b}$ का मान होगा : [JEE-2011]

Ans. 9

Sol. $\vec{a} = -\hat{i} - \hat{k}$

$$\vec{b} = -\hat{i} + \hat{j}$$

$$\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$$

$$\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$$

Taking cross product by $\vec{a}$

$$(\vec{r} \times \vec{b}) \times \vec{a} = (\vec{c} \times \vec{b}) \times \vec{a}$$

$$\Rightarrow (\vec{r} \cdot \vec{a})\vec{b} - (\vec{b} \cdot \vec{a})\vec{r} = (\vec{c} \cdot \vec{a})\vec{b} - (\vec{b} \cdot \vec{a})\vec{c}$$

$$\Rightarrow 0 - \vec{r} = (-1 - 3)(-\hat{i} + \hat{j}) - (1)(\hat{i} + 2\hat{j} + 3\hat{k})$$

$$\vec{r} = -3\hat{i} + 6\hat{j} + 3\hat{k}$$

$$\vec{r} \cdot \vec{b} = 3 + 6 = 9$$

STP AND ITS APPLICATION

20. Let the volume of a tetrahedron having the vertices as $\vec{0}, \vec{a}, \vec{b}$ and $\vec{c}$ is λ_1 (unit cubes) and volume of another tetrahedron formed by the centroids of all faces of given tetrahedron is λ_2 (in unit cubes), then $\frac{\lambda_1}{3\lambda_2}$ is.

माना चतुष्फलक जिसके शीर्ष $\vec{0}, \vec{a}, \vec{b}$ तथा $\vec{c}$ है, का आयतन λ_1 (घन इकाई में) तथा एक दूसरा चतुष्फलक जो दिये गये चतुष्फलक के फलकों के केन्द्रों से मिलकर बना है, का आयतन λ_2 (घन इकाई में) है तो $\frac{\lambda_1}{3\lambda_2}$ का मान होगा।

Ans. 9

Sol. Volume $\Rightarrow$

$$\lambda_1 = \frac{1}{6} [\vec{a} \vec{b} \vec{c}]$$

$$\lambda_2 = \frac{1}{6} \left[\frac{\vec{a}}{3} \frac{\vec{b}}{3} \frac{\vec{c}}{3} \right]$$

$$= \frac{1}{6 \times 27} [\vec{a} \vec{b} \vec{c}]$$

$$\frac{\lambda_1}{3\lambda_2} = \frac{\frac{1}{6} [\vec{a} \vec{b} \vec{c}]}{3 \times \frac{1}{6 \times 27} [\vec{a} \vec{b} \vec{c}]} = 9$$

21. Let the volume of tetrahedron ABCD is 81 cubic units and G_1, G_2, G_3 are centroids of triangular faces ABC, ABD and ACD respectively, then volume of tetrahedron $AG_1G_2G_3$ is. (in cubic units)

माना चतुष्फलक ABCD का आयतन 81 घन इकाई तथा G_1, G_2, G_3 क्रमशः ABC, ABD एवं ACD के त्रिकोणीय फलकों के केन्द्रक हो, तो चतुष्फलक $AG_1G_2G_3$ का आयतन होगा। (घन इकाई में)

Ans. 6

Sol. $\frac{1}{6} [\vec{a} \vec{b} \vec{c}] = 81$

$$AG_1G_2G_3$$

$$\begin{aligned}
&= \frac{1}{6} \left[\frac{\vec{a} + \vec{b}}{3} \cdot \frac{\vec{b} + \vec{c}}{3} \cdot \frac{\vec{c} + \vec{a}}{3} \right] \\
&= \frac{1}{27 \times 6} \times 2 [\vec{a} \cdot \vec{b} \cdot \vec{c}] \\
&= 6
\end{aligned}$$

VTP AND ITS APPLICATION

22. If $\vec{a}$ and $\vec{b}$ are vectors in space given by $\vec{a} = \frac{\hat{i} - 2\hat{j}}{\sqrt{5}}$ and $\vec{b} = \frac{2\hat{i} + \hat{j} + 3\hat{k}}{\sqrt{14}}$, then the value of $(2\vec{a} + \vec{b}) \cdot [(\vec{a} \times \vec{b}) \times (\vec{a} - 2\vec{b})]$, is.

यदि $\vec{a}$ तथा $\vec{b}$ इस प्रकार है कि $\vec{a} = \frac{\hat{i} - 2\hat{j}}{\sqrt{5}}$ तथा $\vec{b} = \frac{2\hat{i} + \hat{j} + 3\hat{k}}{\sqrt{14}}$ है, तो $(2\vec{a} + \vec{b}) \cdot [(\vec{a} \times \vec{b}) \times (\vec{a} - 2\vec{b})]$ का मान होगा।

Ans. 5

Sol. $\vec{a} = \frac{1}{\sqrt{5}}(1, -2, 0)$

$$\vec{b} = \frac{1}{\sqrt{14}}(2, 1, 3)$$

$$(2\vec{a} + \vec{b}) \cdot [(\vec{a} \times \vec{b}) \times (\vec{a} - 2\vec{b})]$$

$$[2\vec{a} + \vec{b}] \cdot [\vec{b}(\vec{a}^2 - 2\vec{a} \cdot \vec{b}) - \vec{a}(\vec{b} \cdot \vec{a} - 2b^2)]$$

$$[2\vec{a} + \vec{b}] \cdot [\vec{b}(1 - 2 \times 0) - \vec{a}(0 - 2)]$$

$$(2\vec{a} + \vec{b}) \cdot (\vec{b} + 2\vec{a})$$

$$4 + 1 + 0 = 5$$

DIRECTION COSINES AND DIRECTION RATIOS

23. Find the angle between the two straight lines whose direction cosines l, m, n are given by $2l + 2m - n = 0$ and $mn + nl + lm = 0$.

उन दो रेखाओं के बीच का कोण बताओं जिनके दिक् कोज्याओं l, m, n में निम्न $2l + 2m - n = 0$ तथा $mn + nl + lm = 0$ है।

Ans. $\theta = 90^\circ$

Sol. $2l + 2m = n$
 $lm + mn + nl = 0$
 $lm + (m + l)(2m + 2l) = 0$
 $lm + 2m^2 + 2l^2 + 4ml = 0$
 $2m^2 + 2l^2 + 5ml = 0$
 $2m^2 + 4ml + ml + 2l^2 = 0$
 $(2m + l)(m + 2l) = 0$
Case-I
 $2m + l = 0$
 $n = -2m = l$

$$v_1 = \left\langle \mp \frac{2}{3}, \pm \frac{1}{3}, \mp \frac{2}{3} \right\rangle$$

Case-II

$$m + 2l = 0$$

$$m = n = -2l$$

$$v_2 = \left\langle \pm \frac{1}{3}, \mp \frac{2}{3}, \mp \frac{2}{3} \right\rangle$$

$$v_1 \cdot v_2 = -\frac{2}{3} - \frac{2}{3} + \frac{4}{3} = 0$$

EQUATION OF LINE

24. Find out whether the following pairs of lines are parallel, non-parallel and intersecting, or non-parallel and non-intersecting :

ज्ञात करो कि इनमें से कौन सा रेखा युग्म समान्तर, असमान्तर प्रतिच्छेदित एवं विषमतलीय होगा :

$$(i) \quad \begin{aligned} \vec{r}_1 &= \hat{i} + \hat{j} + 2\hat{k} + \lambda (3\hat{i} - 2\hat{j} + 4\hat{k}) \\ \vec{r}_2 &= 2\hat{i} + \hat{j} + 3\hat{k} + \mu (-6\hat{i} + 4\hat{j} - 8\hat{k}) \end{aligned} \quad (ii) \quad \begin{aligned} \vec{r}_1 &= \hat{i} - \hat{j} + 3\hat{k} + \lambda (\hat{i} - \hat{j} + \hat{k}) \\ \vec{r}_2 &= 2\hat{i} + 4\hat{j} + 6\hat{k} + \mu (2\hat{i} + \hat{j} + 3\hat{k}) \end{aligned}$$

$$(iii) \quad \begin{aligned} \vec{r}_1 &= \hat{i} + \hat{k} + \lambda (\hat{i} + 3\hat{j} + 4\hat{k}) \\ \vec{r}_2 &= 2\hat{i} + 3\hat{j} + \mu (4\hat{i} - \hat{j} + \hat{k}) \end{aligned}$$

Ans. (i) parallel (ii) the lines intersect at the point p.v. $-2\hat{i} + 2\hat{j}$ (iii) lines are skew

Sol. (i) $\left. \begin{aligned} (3, -2, 4) \\ (-6, 4, -8) \end{aligned} \right\} \text{Parallel}$

(ii) $\left. \begin{aligned} (1, -1, 1) \\ (2, 1, 3) \end{aligned} \right\} \text{Intersect}$

$$(iii) \left. \begin{matrix} (1, 3, 4) \\ (4, -1, 1) \end{matrix} \right\} \text{Skew}$$

25. Find the equations of the straight line passing through the point $(1, 2, 3)$ to intersect the straight line $x + 1 = 2(y - 2) = z + 4$ and parallel to the plane $x + 5y + 4z = 0$.

बिन्दु $(1, 2, 3)$ से गुजरने वाली तथा रेखा $x + 1 = 2(y - 2) = z + 4$ को प्रतिच्छेदित करने वाली उस रेखा का समीकरण बताओं जो तल $x + 5y + 4z = 0$ के समान्तर हो।

Ans. $\frac{x-1}{2} = \frac{y-2}{2} = \frac{z-3}{-3}$

Sol. $\frac{x-1}{2} = \frac{y-2}{2} = \frac{z-3}{-3}$
 $(2\lambda - 2, \lambda, -7 + 2\lambda) \cdot (1, 5, 4) = 0$
 $2\lambda - 2 + 5\lambda - 28 + 8\lambda = 0$
 $15\lambda - 30 = 0$
 $\lambda = 2$
 $DR's = \langle 2, 2, -3 \rangle$
 $\frac{x-1}{2} = \frac{y-2}{2} = \frac{z-3}{-3}$

26. Find the equations of the two lines through the origin which intersect the line $\frac{x-3}{2} = \frac{y-3}{1} = \frac{z}{1}$ at an angle of $\frac{\pi}{3}$.

मूल बिन्दु से गुजरने वाली उन दोनों रेखाओं का समीकरण बताओं जो रेखा $\frac{x-3}{2} = \frac{y-3}{1} = \frac{z}{1}$ को $\frac{\pi}{3}$ के कोण पर प्रतिच्छेदित करती है।

Ans. $\frac{x}{1} = \frac{y}{2} = \frac{z}{-1}$ or $\frac{x}{-1} = \frac{y}{1} = \frac{z}{-2}$

Sol. Let point on line $\frac{x-3}{2} = \frac{y-3}{1} = \frac{z}{1}$ (1)

are $(3 + 2\lambda, 3 + \lambda, \lambda)$

Equation of line which pass through origin is

$$\frac{x-0}{3+2\lambda} = \frac{y-0}{3+\lambda} = \frac{z-0}{\lambda} \quad \dots\dots(2)$$

Angle between (1) & (2)

$$\cos \frac{\pi}{3} = \frac{(3+2\lambda)2 + (3+\lambda)1 + \lambda \times 1}{\sqrt{(3+2\lambda)^2 + (3+\lambda)^2 + \lambda^2} \sqrt{2^2 + 1^2 + 1}}$$

Solving we get

$$\lambda^2 + 3\lambda + 2 = 0$$

$$\Rightarrow \lambda = -1, -2$$

Putting the value of λ in equation (2)

$$\frac{x}{1} = \frac{y}{2} = \frac{z}{-1} \text{ or } \frac{x}{-1} = \frac{y}{1} = \frac{z}{-2}$$

27. Find the equation to the line passing through the point $(1, -2, -3)$ and parallel to the line $2x + 3y - 3z + 2 = 0 = 3x - 4y + 2z - 4$.

बिन्दु $(1, -2, -3)$ से गुजरने वाली तथा रेखा $2x + 3y - 3z + 2 = 0 = 3x - 4y + 2z - 4$ के समानान्तर रेखा का समीकरण ज्ञात कीजिए।

Ans. $\frac{x-1}{6} = \frac{y+2}{13} = \frac{z+3}{17}$

Sol. DR's of line = $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -3 \\ 3 & -4 & 2 \end{vmatrix} = \hat{i}(-6) - \hat{j}(13) + \hat{k}(-17)$

$$= (6, 13, 17)$$

Line passing from $(1, -2, -3)$ in the direction $= (6, 13, 17)$

$$\frac{x-1}{6} = \frac{y+2}{13} = \frac{z+3}{17}$$

28. Find the equation of the line passing through the point $(4, -14, 4)$ and intersecting the line of intersection of the planes $3x + 2y - z = 5$ and $x - 2y - 2z = -1$ at right angles.

बिन्दु $(4, -14, 4)$ से गुजरने वाली तथा तलों $3x + 2y - z = 5$ तथा $x - 2y - 2z = -1$ के प्रतिच्छेदी रेखा को समकोण पर काटने वाली रेखा का समीकरण ज्ञात कीजिए।

Ans. $\frac{x-4}{3} = \frac{y+14}{10} = \frac{z-4}{4}$

Sol. DR's of line of intersection of planes = $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & -1 \\ 1 & -2 & -2 \end{vmatrix} = \hat{i}(-4-2) - \hat{j}(-6+1) + \hat{k}(-6-2)$

DR's of line of intersection of planes = $(-6, -5, -8)$

For the point on the intersection line of planes let $z = 0$

$$3x + 2y = 5$$

$$x - 2y = -1$$

From these $x = 1, y = 1$

equation of line of intersection

$$\frac{x-1}{6} = \frac{y-1}{-5} = \frac{z}{8}$$

Variable point on the intersection line of planes $P(1 + 6\lambda, 1 - 5\lambda, 8\lambda)$

DR of line joining point $(4, -14, 4)$ & $P(1 + 6\lambda, 1 - 5\lambda, 8\lambda) = (-3 + 6\lambda, 15 - 5\lambda, -4 + 8\lambda)$

$$(-6, 5, -8) \cdot (-3 + 6\lambda, 15 - 5\lambda, -4 + 8\lambda)$$

$$18 - 36\lambda + 75 - 25\lambda + 32 - 64\lambda = 0$$

$$+125 - 125\lambda = 0$$

$$\lambda = 1$$

DR of the line $(3, 10, 4)$

$$\frac{x-4}{3} = \frac{y+14}{10} = \frac{z-4}{4}$$

29. Find the equations to the line which can be drawn from the point $(2, -1, 3)$ perpendicular to the lines

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} \text{ and } \frac{x-4}{4} = \frac{y}{5} = \frac{z+3}{3} \text{ at right angles.}$$

बिन्दु $(2, -1, 3)$ से उन लम्बवत् रेखाओं का समीकरण ज्ञात कीजिए जो रेखाओं $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ तथा $\frac{x-4}{4} = \frac{y}{5} = \frac{z+3}{3}$

के लम्बवत् हो।

Ans. $\frac{x-2}{11} = \frac{y+1}{-10} = \frac{z-3}{2}$

Sol. DR's for perpendicular line = $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 4 & 5 & 3 \end{vmatrix} = \hat{i}(-11) - \hat{j}(-10) + \hat{k}(-2)$

$$\text{line } \frac{x-2}{11} = \frac{y+1}{-10} = \frac{z-3}{2}$$

30. Find the value of p so that the lines $\frac{x+1}{-3} = \frac{y-p}{2} = \frac{z+2}{1}$ and $\frac{x}{1} = \frac{y-7}{-3} = \frac{z+7}{2}$ are in the same plane. For this value of p , find the coordinates of their point of intersection and the equation of the plane containing them.

p का मान ज्ञात करो यदि रेखाएं $\frac{x+1}{-3} = \frac{y-p}{2} = \frac{z+2}{1}$ तथा $\frac{x}{1} = \frac{y-7}{-3} = \frac{z+7}{2}$ एक तल में हो। p के इस मान के लिये इनका प्रतिच्छेद बिन्दु तथा इनको समाहित करने वाले तल का समीकरण ज्ञात कीजिए।

Ans. $p = 3, (2, 1, -3); x + y + z = 0$

Sol.
$$\begin{vmatrix} -3 & 2 & 1 \\ 1 & -3 & 2 \\ +1 & 7-P & -5 \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} -3 & 2 & 1 \\ 1 & -3 & 2 \\ 0 & 10-P & -7 \end{vmatrix} = 0$$

$$+3(21 - 20 + 2P) = 0$$

$$+1(-14 - 10 + P)$$

$$3 + 6P - 24 + P = 0$$

$$P = 3$$

$$\text{Equation of plane} = \begin{vmatrix} x & y-7 & z+7 \\ -3 & 2 & 1 \\ 1 & -3 & 2 \end{vmatrix} = 0$$

$$p = 3, (2, 1, -3); x + y + z = 0$$

PLANE

31. Let $P = (1, 0, -1); Q = (1, 1, 1)$ and $R = (2, 1, 3)$ are three points :

- Find the area of the triangle having P, Q and R as its vertices.
- Give the equation of the plane through P, Q and R in the form $ax + by + cz = 1$.
- Where does the plane in part (b) intersect the y -axis.
- Give parametric equations for the line through R that is perpendicular to the plane in part (b).

यदि $P = (1, 0, -1); Q = (1, 1, 1)$ तथा $R = (2, 1, 3)$ तीन बिन्दु है तो :

- त्रिभुज PQR का क्षेत्रफल ज्ञात कीजिए।
- P, Q तथा R से जाने वाले तल का समीकरण $ax + by + cz = 1$ के रूप में ज्ञात कीजिए।
- खण्ड (b) में तल y -अक्ष पर किस बिन्दु पर प्रतिच्छेदित करेगा।
- R से गुजरने वाली उस रेखा का प्राचल समीकरण ज्ञात करो जो खण्ड (b) में दिये गये तल के लम्बवत् हो।

Ans. (a) $\frac{3}{2}$

(b) $\frac{2x}{3} + \frac{2y}{3} - \frac{z}{3} = 1$

(c) $\left(0, \frac{3}{2}, 0\right)$

(d) $x = 2t + 2; y = 2t + 1$ and $z = -t + 3$

Sol. (i) $\frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -1 & -2 \\ 1 & 0 & 2 \end{vmatrix}$

$$\frac{1}{2} |\hat{i}(-2) - \hat{j}(+2) + \hat{k}(+1)|$$

$$= \frac{1}{2} 3 = \frac{3}{2}$$

(ii) $\vec{n} = (-2, -2, 1)$

$$2x + 2y - z = 3$$

(iii) y-axis $0, \frac{3}{2}$

(iv) $\frac{x-2}{2} = \frac{y-1}{2} = \frac{z-3}{-1} = \lambda$

$$x = 2\lambda + 2 \quad y = 2\lambda + 1 \quad z = -\lambda + 3$$

32. Find the point where the line of intersection of the planes $x - 2y + z = 1$ and $x + 2y - 2z = 5$, intersects the plane $2x + 2y + z + 6 = 0$.

उस बिन्दु को ज्ञात करो जहाँ पर तलों $x - 2y + z = 1$ तथा $x + 2y - 2z = 5$ की प्रतिच्छेदी रेखा तल $2x + 2y + z + 6 = 0$ को प्रतिच्छेदित करती है।

Ans. $(1, -2, -4)$

Sol. Planes are $x - 2y + z = 1$ (i)

$$x + 2y - 2z = 5 \quad \text{....(ii)}$$

$$2x + 2y + z = -6 \quad \text{....(iii)}$$

Add (i) + (ii) + (iii)

$$4x + 2y = 0 \Rightarrow y = -2x \quad \text{....(iv)}$$

From equations (iii) - (i)

$$x + 4y = -7 \quad \text{....(v)}$$

from (iv) and (v) we get

$$x = 1, y = -2$$

Put in (i) we get $z = -4$

So point of intersection is $(1, -2, -4)$

33. Feet of the perpendicular drawn from the point $P(2, 3, -5)$ on the axes of coordinates are A, B and C. Find the equation of the plane passing through their feet and the area of $\triangle ABC$.

बिन्दु P (2, 3, -5) से निर्देशी अक्षों पर डाले गये लम्बों के पाद A, B तथा C है। इन पादों से गुजरने तल का समीकरण तथा त्रिभुज ΔABC का क्षेत्रफल ज्ञात कीजिए।

Ans. $\frac{x}{2} + \frac{y}{3} + \frac{z}{-5} = 1$, Area = $\frac{19}{2}$ sq. units

Sol. $\frac{x}{2} + \frac{y}{3} + \frac{z}{-5} = 0$

$$\frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -3 & 0 \\ 0 & 3 & 5 \end{vmatrix} = \frac{1}{2} [\hat{i}(-15) - \hat{j}(10) + \hat{k}(6)]$$

$$= \frac{1}{2} \sqrt{225 + 100 + 36}$$

$$= \frac{19}{2}$$

34. Find the equation of the plane containing the straight line $\frac{x-1}{2} = \frac{y+2}{-3} = \frac{z}{5}$ and perpendicular to the plane $x - y + z + 2 = 0$.

रेखा $\frac{x-1}{2} = \frac{y+2}{-3} = \frac{z}{5}$ को समाहित करने वाले उस तल का समीकरण ज्ञात करो जो तल $x - y + z + 2 = 0$ के लम्बवत् हो।

Ans. $2x + 3y + z + 4 = 0$

Sol. Line : $\frac{x-1}{2} = \frac{y+2}{-3} = \frac{z}{5}$

Plane : $x - y + z + 2 = 0$

The vector perpendicular to required plane is

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -3 & 5 \\ 1 & -1 & 2 \end{vmatrix} = 2\hat{i} + 3\hat{j} + \hat{k}$$

Now equation of plane passing through (1, -2, 0)

and perpendicular to $2\hat{i} + 3\hat{j} + \hat{k}$

$$(x-1)2 + (y+2)3 + (z-0)1 = 0$$

$$\Rightarrow 2x + 3y + z + 4 = 0$$

35. Let P_1 be a plane passing through two points $A(1, 2, 0)$ and $B(2, -1, 1)$ and perpendicular to $\vec{r} \cdot (\vec{i} + \vec{j} - 3\vec{k}) = 1$. Let L be a line through $(1, 1, 1)$ and intersect line of intersection of planes P_1 and P_2 perpendicularly where equation of plane P_2 is $\vec{r} \cdot (\vec{i} + \vec{j} + \vec{k}) = 5$. If the equation of line L is $\frac{x+1}{-1} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$, then find the value of $\left(\frac{2y_1 + z_1}{bc} \right)$.

माना P_1 एक तल है जो $A(1, 2, 0)$ तथा $B(2, -1, 1)$ से गुजरता है एवं तल $\vec{r} \cdot (\vec{i} + \vec{j} - 3\vec{k}) = 1$ के लम्बवत् है। L एक रेखा है जो $(1, 1, 1)$ से गुजरती है तथा समतल P_1 तथा P_2 की प्रतिच्छेदी रेखा को लम्बवत् प्रतिच्छेद करती है जहाँ तल P_2 का समीकरण $\vec{r} \cdot (\vec{i} + \vec{j} + \vec{k}) = 5$ है। यदि रेखा L का समीकरण $\frac{x+1}{-1} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$ है तो $\left(\frac{2y_1 + z_1}{bc} \right)$ का मान ज्ञात कीजिये।

Ans. 9

Sol. Equation of plane P

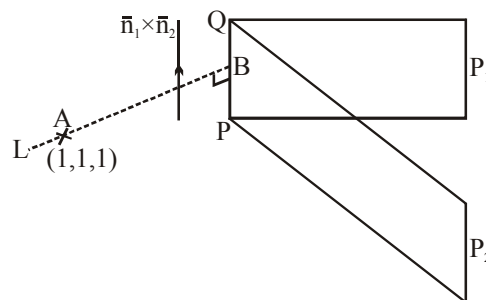
$$\vec{n}_1 = \vec{n} \times \overrightarrow{AB} = (\vec{i} + \vec{j} - 3\vec{k}) \times (\vec{i} - 3\vec{j} + \vec{k}) = 2\vec{i} + \vec{j} + \vec{k}$$

$$\therefore \text{Equation of plane } P_1 : 2(x-1) + 1(y-2) + 1(z-0) = 0 \Rightarrow P_1 : 2x + y + z = 4$$

$$\text{Also, } P_2 : x + y + z = 5.$$

Line PQ is parallel to vector

$$\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{vmatrix} = \vec{k} - \vec{j}$$



$$\therefore \text{Equation PQ: } \frac{x+1}{0} = \frac{y-6}{-1} = \frac{z-0}{1} = \lambda$$

$$\text{Let } B(-1, 6-\lambda, \lambda) \text{ be any point on line PQ. So, } \overrightarrow{AB} = -2\hat{i} + (5-\lambda)\hat{j} + (\lambda-1)\hat{k}$$

$$\text{Now, } \overrightarrow{AB} \cdot \vec{n}_1 \times \vec{n}_2 = 0 \Rightarrow 0 - 5 + \lambda + \lambda - 1 = 0 \Rightarrow \lambda = 3$$

$$\therefore B \equiv (-1, 3, 3)$$

$$\text{So, equation of line L is } \frac{x+1}{1+1} = \frac{y-3}{1-3} = \frac{z-3}{1-3}$$

$$\text{i.e., } \frac{x+1}{-1} = \frac{y-3}{1} = \frac{z-3}{1}$$

$$\text{Hence } \left(\frac{2y_1 + z_1}{bc} = \frac{6+3}{1} \right) = 9. \text{ Ans.}$$

36. Let L be the line given by $\vec{r} = \begin{bmatrix} 2 \\ -2 \\ -1 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ and let P be the point (2, -1, 1). Also suppose that E be the plane

containing three non collinear points A = (0, 1, 1); B(1, 2, 2) and C = (1, 0, 1)

Find

- Distance between the point P and the line L.
- Equation of the plane E.
- Equation the plane F containing the line L and the point P.
- Acute between the plane E and F.
- Volume of the parallelopiped by A, B, C and the point D(-3, 0, 1).

रेखा L का समीकरण है $\vec{r} = \begin{bmatrix} 2 \\ -2 \\ -1 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ तथा बिन्दु P के निर्देशांक (2, -1, 1) है। यदि E एक ऐसा तल है जिस पर तीन

असंरेखीय बिन्दु A = (0, 1, 1); B(1, 2, 2) तथा C = (1, 0, 1) हो तो

ज्ञात कीजिए

- P तथा L के बीच दूरी बताइये।
- तल E का समीकरण ज्ञात कीजिए।
- तल F का समीकरण, जिसमें रेखा L हो तथा बिन्दु P हो, ज्ञात कीजिए।
- तलों E तथा F के बीच न्यून कोण ज्ञात कीजिए।
- A, B, C तथा D(-3, 0, 1) से बने हुए समान्तर षट्फलक का आयतन ज्ञात कीजिये।

Ans. (a) $\sqrt{3}$

(b) $x + y - 2z + 1 = 0$

(c) $x - 2y + z = 5$

(d) $\frac{\pi}{3}$

(e) 4

Sol. A(0,1,1) B(1,2,2) C(1,0,1)

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & -1 & 0 \end{vmatrix} = \hat{i} - \hat{j}(-1) + \hat{k}(-2)$$

(1, 1, -2)

(i) $(-\lambda, -1, -2 + \lambda) \cdot (-1, 0, 1) = 0$

$$\lambda - 2 + \lambda = 0$$

$$\lambda = 1$$

$$\sqrt{1+1+1} = \sqrt{3}$$

$$(ii) x + y - 2z = -1$$

$$(iii) \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 0 & 1 \\ 0 & -1 & -2 \end{vmatrix} = \hat{i} - \hat{j}(2) + \hat{k}$$

$$x - 2y + z = 5$$

$$(iv) = \frac{(1, 1, -2) \cdot (1, -2, 1)}{6}$$

$$= \frac{1 - 2 - 2}{6} = \frac{1}{2}$$

$$\theta = 60^\circ$$

$$(v) \frac{1}{6} \begin{vmatrix} 0 & 2 & 1 \\ -4 & 0 & 0 \\ -1 & 1 & 0 \end{vmatrix}$$

$$(+4(-1)) = 4$$

37. Find the foot and hence the length of the perpendicular from the point (5, 7, 3) to the line $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$.

Also find the equation of the plane in which the perpendicular and the given straight line lie.

बिन्दु (5, 7, 3) से रेखा $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$ पर लम्बपाद तथा लम्ब की लम्बाई ज्ञात करो। उस तल का समीकरण ज्ञात करो

जिसमें दिया गया लम्ब तथा दी गयी रेखा आती है।

Ans. (9, 13, 15) ; 14 ; $9x - 4y - z = 14$

Sol. $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$ L

DR's of perpendicular line from (5, 7, 3)

$(3\lambda + 10, 8\lambda + 22, -5\lambda + 2)$

Now $9\lambda + 30 + 64\lambda + 176 + 25\lambda - 10 = 0$

$98\lambda + 196 = 0 \Rightarrow \lambda = -2$

foot perpendicular (x, y, z) = (9, 13, 15)

perpendicular line $\frac{x-5}{4} = \frac{y-7}{6} = \frac{z-3}{12}$ L₂

Equation of plane having line L₁ & L₂

$$\begin{vmatrix} x-5 & y-7 & z-3 \\ 2 & 3 & 6 \\ 3 & 8 & -5 \end{vmatrix} = 0$$

$$-9(x-5) + 4(y-7) + z - 3 = 0$$

$$\Rightarrow -9x + 4y + z + 14 = 0$$

38. Let the equation of the plane containing the line $x - y - z - 4 = 0 = x + y + 2z - 4$ and is parallel to the line of intersection of the planes $2x + 3y + z = 1$ and $x + 3y + 2z = 2$ be $x + Ay + Bz + C = 0$. Compute the value of $|A + B + C|$.

माना रेखा $x - y - z - 4 = 0 = x + y + 2z - 4$ को समाहित करने वाला तल जो $2x + 3y + z = 1$ तथा $x + 3y + 2z = 2$ की प्रतिच्छेदी रेखा के समान्तर है, का समीकरण $x + Ay + Bz + C = 0$ है तो $|A + B + C|$ का मान ज्ञात कीजिये।

Ans. 11

Sol. $(x - y - z - 4) + \lambda(x + y + 2z - 4) = 0$
 $(1 + \lambda)x - (1 - \lambda)y - (1 - 2\lambda)z - 4(1 + \lambda) = 0$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 1 \\ 1 & 3 & 2 \end{vmatrix} = \hat{i}(3) - \hat{j}(3) + \hat{k}(3)$$

$$(1, -1, 1)$$

$$1 + \lambda + 1 - \lambda - 1 + 2\lambda = 0$$

$$1 + 2\lambda = 0$$

$$\lambda = -\frac{1}{2}$$

$$\frac{1}{2}x - \frac{3}{2}y - 2z - 4 = 0$$

$$x - 3y - 4z - 4 = 0$$

$$x + Ay + Bz + C = 0$$

$$\text{From this } A = -3, B = -4, C = -4$$

39. Find the equation of the plane containing the line $\frac{x-1}{2} = \frac{y}{3} = \frac{z}{2}$ and parallel to the line $\frac{x-3}{2} = \frac{y}{5} = \frac{z-2}{4}$. Find also the S.D. between the two lines.

रेखा $\frac{x-1}{2} = \frac{y}{3} = \frac{z}{2}$ को समाहित करने वाली, तथा $\frac{x-3}{2} = \frac{y}{5} = \frac{z-2}{4}$ के समानान्तर तल का समीकरण ज्ञात करो। इन दोनों

रेखाओं के बीच न्यूनतम दूरी भी ज्ञात कीजिए।

Ans. $x - 2y + 2z - 1 = 0$; 2 units

Sol. DR's of vector perpendicular to both lines = $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 2 \\ 2 & 5 & 4 \end{vmatrix} = \hat{i}(2) - \hat{j}(4) + \hat{k}(4)$

DR's of vector perpendicular to both lines = $\lambda(1, -2, 2)$

Equation of plane

$$x - 2y + 2z = 1$$

$$\text{Shortest distance between lines} = (2, 0, 2) \cdot \frac{(1, -2, 2)}{3} = \frac{2+4}{3} = 2$$

40. Consider the plane

$$E: \vec{r} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + \mu \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Let F be the plane containing the point A(-4, 2, 2) and parallel to E.

Suppose the point B is on the plane E such that B has a minimum distance from the point A. If C(-3, 0, 4) lies in the plane F. Find the area of the triangle ABC.

माना समतल

$$E: \vec{r} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + \mu \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

F बिन्दु A(-4, 2, 2) से गुजरने वाला E के समान्तर तल है। यदि बिन्दु B तल E में स्थित हो तथा A से न्यूनतम दूरी पर तो तथा C(-3, 0, 4) तल F में हो तो त्रिभुज ABC का क्षेत्रफल ज्ञात कीजिए।

Ans. $\frac{9}{2}$

Sol. $E: \vec{r} = (-1, 1, 1) + \lambda(1, 2, 0) + \mu(1, 0, 1) \rightarrow B$

$$F \rightarrow A(-4, 2, 2)$$

$$BA \Rightarrow \min$$

$$C(-3, 0, 4)$$

$$E = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 0 \\ 1 & 0 & 1 \end{vmatrix} = \hat{i}(2) - \hat{j}(1) + \hat{k}(-2) \Rightarrow E = 2x - y - 2z = -5$$

$$= (2, -1, -2)$$

$$A \rightarrow FOP$$

$$(-4 + 2\lambda, 2 - \lambda, 2 - 2\lambda)$$

$$B(-2, 1, 0)$$

$$-8 + 4\lambda - 2 + \lambda - 4 + 4\lambda = -5$$

$$9\lambda = 9$$

$$\lambda = 1$$

$$\frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 1 & 2 \\ -1 & 2 & -2 \end{vmatrix}$$

$$= \frac{1}{2} |-6\hat{i} + 6\hat{j} - 3\hat{k}|$$

$$= \frac{9}{2}$$

41. Let P be the plane passing through (1, 1, 1) and parallel to the lines L_1 and L_2 having direction ratios 1, 0, -1 and -1, 1, 0 respectively. If A, B and C are the points at which P intersects the coordinate axes, find the volume of the tetrahedron whose vertices are A, B, C and the origin.

माना P एक तल है जो (1, 1, 1) से गुजरता है तथा रेखाएँ L_1 तथा L_2 जिनके दिक् अनुपात 1, 0, -1 तथा -1, 1, 0 हैं, के समान्तर है। यदि A, B तथा C वह बिन्दु है जहाँ P निर्देशी अक्षों को काटता है तो चतुष्फलक का आयतन ज्ञात कीजिये जिसके शीर्ष A, B, C तथा मूल बिन्दु है।

Ans. $\frac{9}{2}$

Sol. $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{vmatrix} = \hat{i} + \hat{j} + \hat{k}(\cdot)$ $x + y + z = 3$

$$\frac{1}{6} \begin{vmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{vmatrix} = \frac{1}{6} \times 27$$

$$= \frac{9}{2}$$

FAMILY OF PLANES

42. Let (x, y, z) be points with integer coordinates satisfying the system of homogeneous equations

$$3x - y - z = 0 ; -3x + z = 0 ; -3x + 2y + z = 0$$

Then the number of such points for which $x^2 + y^2 + z^2 \leq 100$ is.

माना (x, y, z) बिन्दु है जिनके निर्देशांक पूर्णांक हैं तथा निम्न समघातीय समीकरणों को सन्तुष्ट करते हैं

$$3x - y - z = 0 ; -3x + z = 0 ; -3x + 2y + z = 0$$

तो ऐसे बिन्दुओं की संख्या होगी जिनके लिये $x^2 + y^2 + z^2 \leq 100$ हो।

Ans. 7

Sol. $3x - y - z = 0$
 $-3x + z = 0$
 $-3x + 2y + z = 0$
 $y = 0$
 $z = 3x$
 $x^2 + 0 + 9x^2 \leq 100$
 $10x^2 \leq 100$
 $x^2 \leq 10$
 $x = 0, \pm 1, \pm 2, \pm 3$
 $= 7$

43. If the distance between the plane $Ax - 2y + z = d$ and the plane containing the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ is $\sqrt{6}$, then $|d|$ is.

यदि तल $Ax - 2y + z = d$ तथा रेखाएँ $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ तथा $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ को समाहित करने वाले तल के मध्य दूरी $\sqrt{6}$ है तो $|d|$ का मान होगा।

Ans. 6

Sol. $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 1 & 1 & 1 \end{vmatrix} = \hat{i}(-1) - \hat{j}(-2) + \hat{k}(-1)$
 $(1, -2, 1)$
 $x - 2y + z = d$
 $x - 2y + z = 0$
 $\left| \frac{d}{\sqrt{6}} \right| = \sqrt{6}$
 $|d| = 6$

44. The plane denoted by $\Pi_1 : 4x + 7y + 4z + 81 = 0$ is rotated through a right angle about its line of intersection with the plane $\Pi_2 : 5x + 3y + 10z = 25$. If the plane in its new position be denoted by Π , and the distance of this plane from the origin is $\sqrt{k}$ where $k \in \mathbb{N}$, then find k .

तल $\Pi_1 : 4x + 7y + 4z + 81 = 0$ को इसके तथा तल $\Pi_2 : 5x + 3y + 10z = 25$ के प्रतिच्छेदी रेखा के सापेक्ष समकोण पर घुमाया जाता है। यदि तल का नयी स्थिति Π हो तथा इस तल का मूल बिन्दु से दूरी $\sqrt{k}$, $k \in \mathbb{N}$ हो तो k का मान ज्ञात कीजिए।

Ans. 212

Sol. $\pi_1 : 4x + 7y + 4z + 81 = 0$
 $\pi_2 : 5x + 3y + 10z = 25$
 $\pi_1 + \lambda\pi_2 = 0$
 $(4 + 5\lambda)x + (7 + 3\lambda)y + (4 + 10\lambda)z + (81 - 25\lambda) = 0$
 $(4, 7, 4) = 0$
 $16 + 20\lambda + 49 + 21\lambda + 16 + 40\lambda = 0$
 $81\lambda + 81 = 0$
 $\lambda = -1$
 $-x + 4y - 6z + 106 = 0$
 $\frac{106}{\sqrt{16+1+36}} = \frac{106}{\sqrt{53}} = 2\sqrt{53} = \sqrt{212}$

45. Find the equation of the plane containing the line $2x - y + z - 3 = 0$, $3x + y + z = 5$ and at a distance of $\frac{1}{\sqrt{6}}$ from the point $(2, 1, -1)$.

उस समतल का समीकरण ज्ञात कीजिये, जिस पर रेखा $2x - y + z - 3 = 0$, $3x + y + z = 5$ स्थित है तथा जिसकी बिन्दु $(2, 1, -1)$ से दूरी $\frac{1}{\sqrt{6}}$ है।

Ans. $2x - y + z - 3 = 0$ and $62x + 29y + 19z - 105 = 0$

Sol. Equation of plane containing the line,
 $2x - y + z - 3 = 0$ and $3x + y + z = 5$ is
 $(2x - y + z - 3) + \lambda(3x + y + z - 5) = 0$
 $\Rightarrow (2+3\lambda)x + (\lambda-1)y + (\lambda+1)z - 3 - 5\lambda = 0$

Since distance of plane from $(2, 1, -1)$ to above plane is $\frac{1}{\sqrt{6}}$

$$\therefore \left| \frac{6\lambda + 4 + \lambda - 1 - \lambda - 1 - 3 - 5\lambda}{\sqrt{(3\lambda+2)^2 + (\lambda-1)^2 + (\lambda+1)^2}} \right| = \frac{1}{\sqrt{6}}$$

$$\Rightarrow 6(\lambda - 1)^2 = 11\lambda^2 + 12\lambda + 6$$

$$\Rightarrow \lambda = 0, -\frac{24}{5}$$

$\therefore$ Equation of planes are,
 $2x - y + z - 3 = 0$ and $62x + 29y + 19z - 105 = 0$

CONCEPT OF IMAGE

46. Find the equation of the line which is reflection of the line $\frac{x-1}{9} = \frac{y-2}{-1} = \frac{z+3}{-3}$ in the plane $3x - 3y + 10z = 26$.

तल $3x - 3y + 10z = 26$ में रेखा $\frac{x-1}{9} = \frac{y-2}{-1} = \frac{z+3}{-3}$ का प्रतिबिम्ब ज्ञात कीजिए।

Ans. $\frac{x-4}{9} = \frac{y+1}{-1} = \frac{z-7}{-3}$

Sol. Line parallel to plane
Reflection of line will be parallel to given line
mirror image of point $P(1, 2, -3)$
FOP from P in the direction $(3, -3, 10)$ will be $(1 + 3\lambda, 2 - 3\lambda, -3 + 10\lambda)$
FOP on plane
 $3 + 9\lambda - 6 + 9\lambda - 30 + 100\lambda = 26$

$$\lambda = \frac{1}{2}$$

$$\text{FOP} \left(\frac{5}{2}, \frac{1}{2}, 2 \right)$$

Mirror image of point P $(4, -1, 7)$

$$\text{Equation of reflected line } \frac{x-4}{9} = \frac{y+1}{-1} = \frac{z-7}{-3}$$

MISCELLANEOUS

47. Find the equations to the line of greatest slope through the point $(7, 2, -1)$ in the plane $x - 2y + 3z = 0$ assuming that the axes are so placed that the plane $2x + 3y - 4z = 0$ is horizontal.

तल $x - 2y + 3z = 0$ में $(7, 2, -1)$ में महत्तम ढाल वाली रेखा का समीकरण ज्ञात करो। अक्षों को इस तरह रखा गया है कि तल $2x + 3y - 4z = 0$ क्षैतिज है।

Ans. $\frac{x-7}{22} = \frac{y-2}{5} = \frac{z+1}{-4}$

Sol. Line is $\perp$ to intersection line of planes

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -4 \\ 1 & -2 & 3 \end{vmatrix} = \hat{i} - \hat{j}10 + \hat{k}(-7) = (1, -10, -7)$$

direction of line

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -10 & -7 \\ 1 & -2 & 3 \end{vmatrix} = \hat{i}(-44) - \hat{j}(10) + \hat{k}(8)$$

$$\langle 22, 5, -4 \rangle$$

$$\text{Equation of line } \frac{x-7}{22} = \frac{y-2}{5} = \frac{z+1}{-4}$$

Paragraph for Question Nos. 48 to 50

Let $A(1, 2, 3)$, $B(0, 0, 1)$, $C(-1, 1, 1)$ are the vertices of a ΔABC .

माना $A(1, 2, 3)$, $B(0, 0, 1)$, $C(-1, 1, 1)$ त्रिभुज ABC के शीर्ष है

48. The equation of internal angle bisector through A to side BC is :

कोण A के आन्तरिक कोणार्धक का समीकरण है :

(A) $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 2\hat{j} + 3\hat{k})$

(B) $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 4\hat{j} + 3\hat{k})$

(C) $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 3\hat{j} + 2\hat{k})$

(D) $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 3\hat{j} + 4\hat{k})$

Ans. (D)

Sol. Here ΔABC is isosceles with $AB = AC$,
So internal bisector of A is perpendicular to BC .

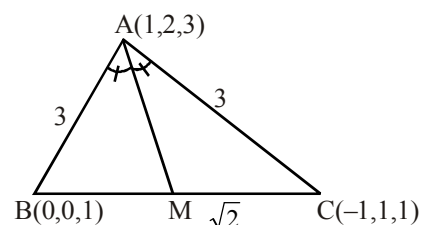
As $\Delta AMB \cong \Delta AMC$ (R.H.S. rule)

$\Rightarrow$ M is midpoint of BC .

$$\text{So, } M \equiv \left(\frac{-1}{2}, \frac{1}{2}, 1 \right)$$

$\therefore$ Equation of internal bisector through A to side BC is

$$\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \mu \left(\frac{3}{2}\hat{i} + \frac{3}{2}\hat{j} + 2\hat{k} \right) \text{ OR } \vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \mu(3\hat{i} + 3\hat{j} + 4\hat{k})$$



Alternative for (i): Equation of BC is

$$\vec{r} = \hat{k} + \lambda(\hat{i} - \hat{j})$$

Let p.v. of M on BC be $\vec{r}$

$$\text{Now, } \vec{AM} = \text{p.v. of } M - \text{p.v. of } A = (\lambda\hat{i} - \lambda\hat{j} + \hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k}) = (\lambda - 1)\hat{i} - (\lambda + 2)\hat{j} - 2\hat{k}$$

$$\text{Since } \vec{AM} \cdot (\hat{i} - \hat{j}) = 0 \quad \Rightarrow \quad \lambda = -1/2$$

$$\therefore \text{ p.v. of point } M = \frac{-1}{2}\hat{i} + \frac{1}{2}\hat{j} + \hat{k}$$

$\Rightarrow$ Equation of internal bisector through A to the side BC is

$$\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \mu \left(\frac{3}{2}\hat{i} + \frac{3}{2}\hat{j} + 2\hat{k} \right) \text{ OR } \vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \mu (3\hat{i} + 3\hat{j} + 4\hat{k})$$

49. The equation of median through C to side AB is :

C से मधिका का समीकरण होगा :

(A) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(3\hat{i} - 2\hat{k})$

(B) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(3\hat{i} + 2\hat{k})$

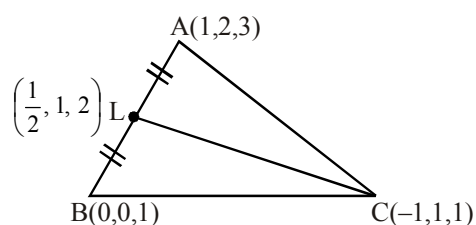
(C) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(-3\hat{i} + 2\hat{k})$

(D) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(3\hat{i} + 2\hat{j})$

Ans. (B)

Sol. Clearly midpoint L of AB is $\left(\frac{1}{2}, 1, 2 \right)$

$\Rightarrow$ Equation of median through C to AB is



$$\vec{r} = (-\hat{i} + \hat{j} + \hat{k}) + p \left(\frac{3}{2}\hat{i} + \hat{k} \right) \text{ OR } \vec{r} = (-\hat{i} + \hat{j} + \hat{k}) + p(3\hat{i} + 2\hat{k})$$

50. The area (ΔABC) is equal to :

ΔABC का क्षेत्रफल है :

(A) $\frac{9}{2}$

(B) $\frac{\sqrt{17}}{2}$

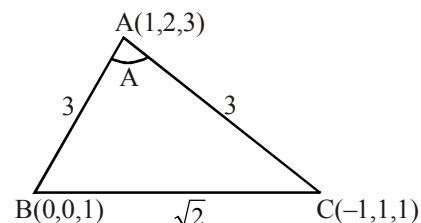
(C) $\frac{17}{2}$

(D) $\frac{7}{2}$

Ans. (B)

Sol. We have $\cos A = \frac{3^2 + 3^2 - (\sqrt{2})^2}{2(3)(3)}$

$$\therefore \cos A = \frac{16}{18} = \frac{8}{9}$$



$$\text{Now area } (\Delta ABC) = \frac{1}{2} (3) (3) \sin A = \frac{9}{2} \sqrt{1 - \frac{64}{81}} = \frac{9}{2} \times \frac{\sqrt{17}}{9} = \frac{\sqrt{17}}{2} \text{ sq. unit.}$$

Paragraph for Question Nos. 51 to 52

Consider a plane $x + y - z = 1$ and the point $A(1, 2, -3)$ है। A line has the equation $x = 1 + 3r, y = 2 - r, z = 3 + 4r$. माना एक तल $x + y - z = 1$ तथा बिन्दु $A(1, 2, -3)$ है। एक रेखा L का समीकरण $x = 1 + 3r, y = 2 - r, z = 3 + 4r$

51. The co-ordinate of a point B of line L, such that AB is parallel to the plane, is :

बिन्दु B निर्देशांक ज्ञात करें जो रेखा L पर स्थित है तथा AB समतल के समानान्तर है :

- (A) 10, -1, 15 (B) -5, 4, -5 (C) 4, 1, 7 (D) -8, 5, -9

Ans. D

Sol. line $\frac{x-1}{3} = \frac{y-2}{-1} = \frac{z-3}{4} = r; A(1, 2, -3)$

Any point say $B \equiv 3r + 1, 2 - r, 3 + 4r$ (on the line L)

$$\overline{AB} = 3r, -r, 4r + 6$$

Hence, $\overline{AB}$ is parallel to $x + y - z = 1$

$$\text{hence, } 3r - r - 4r - 6 = 0$$

$$2r = -6 \quad \Rightarrow \quad r = -3$$

hence B is -8, 5, -9

52. Equation of the plane containing the line L and the point A has the equation :

रेखा L को समाहित करने वाले तथा बिन्दु A से गुजरने वाले समतल का समीकरण है :

- (A) $x - 3y + 5 = 0$ (B) $x + 3y - 7 = 0$ (C) $3x - y - 1 = 0$ (D) $3x + y - 5 = 0$

Ans. B

Sol. Equation of plane containing the line L is

$$A(x - 1) + B(y - 2) + C(z - 3) = 0, \text{ where } 3A - B + 4C = 0 \quad \dots(1)$$

$\therefore$ (1) also contains the point $A(1, 2, -3)$

hence $C = 0; 3A = B$

$$\text{equation of plane} \quad x - 1 + 3(y - 2) = 0$$

$$x + 3y - 7 = 0 \quad \text{Ans.}$$

Paragraph for Question Nos. 53 to 55

$$\text{Consider the lines } L_1: \frac{x+1}{3} = \frac{y+2}{1} = \frac{z+1}{2}; \quad L_2: \frac{x-2}{1} = \frac{y+2}{2} = \frac{z-3}{3}$$

$$\text{माना रेखायें } L_1: \frac{x+1}{3} = \frac{y+2}{1} = \frac{z+1}{2}; \quad L_2: \frac{x-2}{1} = \frac{y+2}{2} = \frac{z-3}{3}$$

53. The unit vector perpendicular to both L_1 and L_2 is :

दोनों रेखाओं L_1 व L_2 के लम्बवत् इकाई सदिश है :

(A) $\frac{-\hat{i}+7\hat{j}+7\hat{k}}{\sqrt{99}}$ (B) $\frac{-\hat{i}-7\hat{j}+5\hat{k}}{5\sqrt{3}}$ (C) $\frac{-\hat{i}+7\hat{j}+5\hat{k}}{5\sqrt{3}}$ (D) $\frac{7\hat{i}-7\hat{j}-\hat{k}}{\sqrt{99}}$

Ans. B

Sol.
$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{vmatrix} = \hat{i}(-1) - \hat{j}(7) + \hat{k}(5)$$

$$= \frac{\langle -1, -7, 5 \rangle}{\sqrt{75}}$$

54. The shortest distance between L_1 and L_2 is :

L_1 तथा L_2 के बीच की न्यूनतम दूरी है :

(A) 0 (B) $\frac{17}{\sqrt{3}}$ (C) $\frac{41}{5\sqrt{3}}$ (D) $\frac{17}{5\sqrt{3}}$

Ans. D

Sol.
$$(\vec{a}_2 - \vec{a}_1) \cdot \frac{(\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|}$$

$$= \langle -3, +0, -4 \rangle \cdot \frac{\langle -1, -7, 5 \rangle}{\sqrt{75}}$$

$$= \frac{|3-20|}{5\sqrt{3}} = \frac{17}{5\sqrt{3}}$$

55. The distance of the point (1, 1, 1) from the plane passing through the point (-1, -2, -1) and whose normal is perpendicular to both the lines L_1 and L_2 is :

बिन्दु (-1, -2, -1) से होकर जाने वाले तल जिसका अभिलम्ब दोनों रेखाओं L_1 व L_2 के लम्बवत् है, से बिन्दु (1, 1, 1) की दूरी है :

(A) $\frac{2}{\sqrt{75}}$ (B) $\frac{7}{\sqrt{75}}$ (C) $\frac{13}{\sqrt{75}}$ (D) $\frac{23}{\sqrt{75}}$

Ans. C

Sol. Plane $\vec{n} = \langle -1, -7, 5 \rangle$ Point $\langle -1, -2, -1 \rangle$
 $-x - 7y + 5z = 10$
 distance from (1, 1, 1)

$$\frac{|-13|}{\sqrt{75}} = \frac{13}{\sqrt{75}}$$

ONE OR MORE THAN ONE CORRECT TYPE

56. A vector of magnitude 10 along the normal to the curve $3x^2 + 8xy + 2y^2 - 3 = 0$ at its point $P(1, 0)$ can be :

वक्र $3x^2 + 8xy + 2y^2 - 3 = 0$ के लम्बवत् इसके बिन्दु $P(1, 0)$ पर दस मापांक का सदिश है :

- (A) $6\hat{i} + 8\hat{j}$ (B) $-6\hat{i} + 8\hat{j}$ (C) $6\hat{i} - 8\hat{j}$ (D) $-6\hat{i} - 8\hat{j}$

Ans. AD

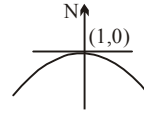
Sol. differentiate the curve

$$6x + 8(xy_1 + y) + 4yy_1 = 0$$

$$m_T \text{ at } (1, 0) \text{ is } 6 + 8(y_1(1)) = 0$$

$$y_1(0) = -\frac{3}{4}$$

$$m_N = \frac{4}{3} \Rightarrow \text{vector along normal } 3\hat{i} + 4\hat{j}$$



$$\text{unit vector} = \pm \frac{(3\hat{i} + 4\hat{j})}{5} \text{ along the normal}$$

again normal vector of magnitude 10 = $\pm (6\hat{i} + 8\hat{j})$ Ans.

57. Consider a plane P passing through $A(\lambda, 3, \mu)$, $B(-1, 3, 2)$ and $C(7, 5, 10)$ and a straight line L with positive direction cosines passing through A, bisecting BC and makes equal angles with the coordinate axes. Let L_1 be a line parallel to L and passing through origin. Which of the following is(are) correct?

(A) The value of $(\lambda + \mu)$ is equal to 5

(B) Equation of straight line L_1 is $\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-1}{1}$

(C) Equation of the plane perpendicular to the plane P and containing line L_1 is $x - 2y + z = 0$

(D) Area of triangle ABC is equal to $3\sqrt{2}$

एक समतल P जो बिन्दुओं $A(\lambda, 3, \mu)$, $B(-1, 3, 2)$ तथा $C(7, 5, 10)$ से गुजरता है। एक रेखा L जिसकी दिक्कोज्याएँ धनात्मक है एवं बिन्दु A से गुजरती है तथा BC को समभाजित करती है एवं समतल निर्देशांक अक्षों के साथ बराबर कोण बनाती है। माना L_1 रेखा L के समानान्तर अन्य रेखा है जो मूल बिन्दु से होकर गुजरती है। निम्न में से कौनसे कथन सही है?

(A) $(\lambda + \mu) = 5$

(B) रेखा L_1 का समीकरण $\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-1}{1}$ है

(C) समतल P के लम्बवत् तल जो रेखा L_1 को समाहित करता है, का समीकरण $x - 2y + z = 0$ है

(D) त्रिभुज ABC का क्षेत्रफल $3\sqrt{2}$

Ans. BCD

Sol. (A) Dr's of Straight L which is passing through A and bisecting BC are $\lambda - 3, -1, \mu - 6$.
Straight line L is equally inclined with co-ordinate axes

$$\therefore \frac{\lambda - 3}{1} = \frac{-1}{1} = \frac{\mu - 6}{1} \Rightarrow \lambda = 2 \text{ and } \mu = 5$$

$$\lambda + \mu = 7. \text{ Ans.}$$

(B) Dr's of line L are 1, 1, 1

$\therefore$ Option 'c' is correct

$$\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-1}{1} \text{ it satisfied } (0, 0, 0).$$

(C) Normal vector of plane P is [12th, 17-10-2010, vector]

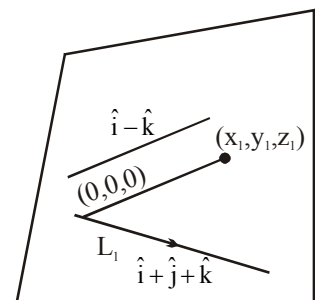
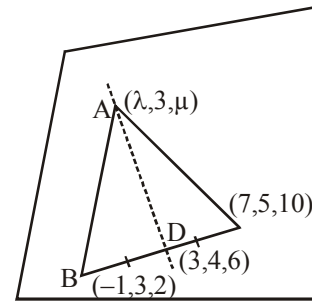
$$\vec{n} = \vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 0 & 3 \\ -5 & -2 & -5 \end{vmatrix} = 6\hat{i} - \hat{j} \times 0 + \hat{k}(-6) = 6(\hat{i} - \hat{k}).$$

Plane perpendicular to plane P and containing line L_1 is

$$= \begin{vmatrix} x & y & z \\ 1 & 0 & -1 \\ 1 & 1 & 1 \end{vmatrix} = 0$$

$$\Rightarrow x(0+1) - y(1+1) + z(1) = 0$$

$$\Rightarrow x - 2y + z = 0. \text{ Ans.}$$



58. A line L passing through the point P(1, 4, 3), is perpendicular to both the lines

$$\frac{x-1}{2} = \frac{y+3}{1} = \frac{z-2}{4} \text{ and } \frac{x+2}{3} = \frac{y-4}{2} = \frac{z+1}{-2}$$

If the position vector of point Q on L is (a_1, a_2, a_3) such that $(PQ)^2 = 357$, then $(a_1 + a_2 + a_3)$ can be :

एक रेखा जो बिन्दु P(1, 4, 3) से गुजरती है तथा रेखाओं

$$\frac{x-1}{2} = \frac{y+3}{1} = \frac{z-2}{4} \text{ तथा } \frac{x+2}{3} = \frac{y-4}{2} = \frac{z+1}{-2}$$

के लम्बवत् है। यदि Q (a_1, a_2, a_3) रेखा पर स्थित है, $(PQ)^2 = 357$, तो $(a_1 + a_2 + a_3)$ हो सकता है :

(A) 16

(B) 15

(C) 2

(D) 1

Ans. BD

Sol. Equation of the line passing through P(1, 4, 3) is

$$\frac{x-1}{a} = \frac{y-4}{b} = \frac{z-3}{c} \dots(1)$$

where $a\hat{i} + b\hat{j} + c\hat{k}$ is along the common perpendicular to both the given lines.

Since (1) is perpendicular to $\frac{x-1}{2} = \frac{y+3}{1} = \frac{z-2}{4}$ and $\frac{x+2}{3} = \frac{y-4}{2} = \frac{z+1}{-2}$

Hence $2a + b + 4c = 0$

and $3a + 2b - 2c = 0$

$$\therefore \frac{a}{-2-8} = \frac{b}{12+4} = \frac{c}{4-3} \Rightarrow \frac{a}{-10} = \frac{b}{16} = \frac{c}{1}$$

Hence the equation of the lines is $\frac{x-1}{-10} = \frac{y-4}{16} = \frac{z-3}{1} \dots(2)$ Ans.

Now any point Q on (2) can be taken as $(1 - 10\lambda, 16\lambda + 4, \lambda + 3)$

$\therefore$ Distance of Q from P $(1, 4, 3) = (10\lambda)^2 + (16\lambda)^2 + \lambda^2 = 357$

$$\Rightarrow (100 + 256 + 1)\lambda^2 = 357$$

$$\Rightarrow \lambda = 1 \text{ or } -1$$

$\therefore$ Q is $(-9, 20, 4)$ or $(11, -12, 2)$

Hence $a_1 + a_2 + a_3 = 15$ or 1 Ans.

ONE OR MORE THAN ONE IS/ARE CORRECT

59. If $\vec{a}$, $\vec{b}$, $\vec{c}$ be three non zero vectors satisfying the condition $\vec{a} \times \vec{b} = \vec{c}$ and $\vec{b} \times \vec{c} = \vec{a}$ then which of the following always hold(s) good?

(A) $\vec{a}$, $\vec{b}$, $\vec{c}$ are orthogonal in pairs

(B) $[\vec{a} \vec{b} \vec{c}] = |\vec{b}|$

(C) $[\vec{a} \vec{b} \vec{c}] = [\vec{a} \vec{b} \vec{c}] = |\vec{c}|^2$

(D) $|\vec{b}| = |\vec{c}|$

यदि $\vec{a}$, $\vec{b}$, $\vec{c}$ तीन अशून्य सदिश इस प्रकार है कि जो शर्त $\vec{a} \times \vec{b} = \vec{c}$ तथा $\vec{b} \times \vec{c} = \vec{a}$ को संतुष्ट करते है तो इनमें से कौनसे कथन सदैव सही होंगे?

(A) $\vec{a}$, $\vec{b}$, $\vec{c}$ युग्म के तौर पर लंबवत् है

(B) $[\vec{a} \vec{b} \vec{c}] = |\vec{b}|$

(C) $[\vec{a} \vec{b} \vec{c}] = [\vec{a} \vec{b} \vec{c}] = |\vec{c}|^2$

(D) $|\vec{b}| = |\vec{c}|$

Ans. AC

Sol. Clearly $\vec{a} \cdot \vec{c} = 0$ and $\vec{b} \cdot \vec{c} = 0$ Also $\vec{a} \cdot \vec{b} = 0 \Rightarrow A$

$$\vec{a} \times \vec{b} = \vec{c}$$

dot with $\vec{b} \Rightarrow \vec{b} \cdot \vec{c} = 0$; $\parallel \text{ly } \vec{b} \times \vec{c} = \vec{a}$

dot with $\vec{b} \Rightarrow \vec{a} \cdot \vec{b} = 0$; dot with $\vec{c} \Rightarrow \vec{a} \cdot \vec{c} = 0$

$$\Rightarrow \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$$

$$\text{Again } \frac{|\vec{a}| |\vec{b}|}{|\vec{b}| |\vec{c}|} = \frac{|\vec{c}|}{|\vec{a}|} \Rightarrow \frac{|\vec{a}|}{|\vec{c}|} = \frac{|\vec{c}|}{|\vec{a}|} \Rightarrow |\vec{a}| = |\vec{c}| \text{ \& } |\vec{b}| = 1$$

$$\Rightarrow \vec{a} \times \vec{b} \cdot \vec{c} = |\vec{a}| |\vec{b}| |\vec{c}| = |\vec{a}|^2 = |\vec{c}|^2$$

(children will assume $\vec{a} = \hat{i}$; $\vec{b} = \hat{j}$ and $\vec{c} = \hat{k}$ but in this case all the four will be correct which will be wrong)

60. Given the following information about the non zero vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$

(i) $(\vec{A} \times \vec{B}) \times \vec{A} = \vec{0}$

(ii) $\vec{B} \cdot \vec{B} = 4$

(iii) $\vec{A} \cdot \vec{B} = -6$

(iv) $\vec{B} \cdot \vec{C} = 6$

Which one of the following holds good?

अशून्य सदिशों $\vec{A}$, $\vec{B}$ तथा $\vec{C}$ के लिये निम्न जानकारी दी गई है :

(i) $(\vec{A} \times \vec{B}) \times \vec{A} = \vec{0}$

(ii) $\vec{B} \cdot \vec{B} = 4$

(iii) $\vec{A} \cdot \vec{B} = -6$

(iv) $\vec{B} \cdot \vec{C} = 6$

निम्न में से कौनसा सही है?

(A) $\vec{A} \times \vec{B} = \vec{0}$

(B) $\vec{A} \cdot (\vec{B} \times \vec{C}) = 0$

(C) $\vec{A} \cdot \vec{A} = 8$

(D) $\vec{A} \cdot \vec{C} = -9$

Ans. ABD

Sol. Given $|\vec{A}| |\vec{B}| \cos \theta = -6$; $|\vec{B}| = 2$ (given)

$$\vec{B} \cdot \vec{C} = |\vec{B}| |\vec{C}| \cos \phi = 6$$

and $(\vec{A} \times \vec{B}) \times \vec{A} = \vec{0}$

$$(\vec{A} \cdot \vec{A})\vec{B} - (\vec{B} \cdot \vec{A})\vec{A} = \vec{0}$$

$$(\vec{A} \cdot \vec{A})\vec{B} = -6\vec{A} \quad \dots(1)$$

$$\therefore \vec{A} \text{ and } \vec{B} \text{ are collinear and } \theta \text{ between } \vec{A} \text{ and } \vec{B} \text{ is } \pi \Rightarrow \vec{A} \times \vec{B} = \vec{0} \Rightarrow A \text{ is correct}$$

$$\Rightarrow \vec{A} \cdot (\vec{B} \times \vec{C}) = (\vec{A} \times \vec{B}) \cdot \vec{C} = 0 \Rightarrow B \text{ is correct}$$

also $\vec{A} \cdot \vec{B} = -6$ and $|\vec{B}| = 2$

$$\therefore |\vec{A}| |\vec{B}| \cos \pi = -6$$

$$|\vec{A}| \cdot (2) = 6 \Rightarrow |\vec{A}| = 3 \Rightarrow \vec{A} \cdot \vec{A} = 9 \Rightarrow C \text{ is not correct}$$

again $\vec{A} \cdot \vec{C} = ?$

dot with $\vec{C}$ the equation (1)

$$9(\vec{B} \cdot \vec{C}) = -6\vec{A} \cdot \vec{C}$$

$$9(6) = -6(\vec{A} \cdot \vec{C}) \Rightarrow \vec{A} \cdot \vec{C} = -9 \Rightarrow \text{(D) is correct}$$

61. If $\vec{A}$, $\vec{B}$, $\vec{C}$ and $\vec{D}$ are four non zero vectors in the same plane no two of which are collinear then which of the following hold(s) good?

यदि $\vec{A}$, $\vec{B}$, $\vec{C}$ तथा $\vec{D}$ चार अशून्य सदिश में स्थित है जिनमें कोई दो समरेखीय नहीं है तो निम्न में से कौनसे कथन सत्य होंगे?

- (A) $(\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = 0$ (B) $(\vec{A} \times \vec{C}) \cdot (\vec{B} \times \vec{D}) \neq 0$
 (C) $(\vec{A} \times \vec{B}) \times (\vec{C} \times \vec{D}) = \vec{0}$ (D) $(\vec{A} \times \vec{C}) \times (\vec{B} \times \vec{D}) \neq \vec{0}$

Ans. BC

Sol. Obviously (B) and (C)

62. If $\vec{a} \times \vec{b} = \vec{c} \times \vec{d}$ and $\vec{a} \times \vec{c} = \vec{b} \times \vec{d}$, then the vectors $\vec{a} - \vec{d}$ and $\vec{b} - \vec{c}$ are :

- (A) collinear (B) linearly independent
 (C) perpendicular (D) parallel

यदि $\vec{a} \times \vec{b} = \vec{c} \times \vec{d}$ और $\vec{a} \times \vec{c} = \vec{b} \times \vec{d}$ हो, तो सदिश $\vec{a} - \vec{d}$ और $\vec{b} - \vec{c}$ है :

- (A) संरेखीय (B) रेखीय रूप से स्वतंत्र
 (C) लम्बवत् (D) समान्तर

Ans. AD

Sol. $\vec{a} \times \vec{b} = \vec{c} \times \vec{d} \quad \dots(i)$

$\vec{a} \times \vec{c} = \vec{b} \times \vec{d} \quad \vec{a} - \vec{d} \text{ \& \& } \vec{b} - \vec{c}$

$\vec{d} \times \vec{b} = \vec{c} \times \vec{a} \quad \dots(ii)$

(i) - (ii)

$(\vec{a} - \vec{d}) \times \vec{b} = \vec{c} \times (\vec{d} - \vec{a})$

$(\vec{a} - \vec{d}) \times (\vec{b} - \vec{c}) = 0$

$\vec{a} - \vec{d} \parallel \vec{b} - \vec{c}$

63. If $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ are linearly independent set of vectors and $K_1\vec{a} + K_2\vec{b} + K_3\vec{c} + K_4\vec{d} = \vec{0}$, then :

- (A) $K_1 + K_2 + K_3 + K_4 = 0$ (B) $K_1 + K_3 = K_2 + K_4 = 0$
 (C) $K_1 + K_4 = K_2 + K_3 = 0$ (D) None of these

यदि $\vec{a}$, $\vec{b}$, $\vec{c}$ एवं $\vec{d}$ रेखीय रूप से स्वतंत्र सदिशों का समुच्चय है और $K_1\vec{a} + K_2\vec{b} + K_3\vec{c} + K_4\vec{d} = 0$ हो, तो :

(A) $K_1 + K_2 + K_3 + K_4 = 0$

(B) $K_1 + K_3 = K_2 + K_4 = 0$

(C) $K_1 + K_4 = K_2 + K_3 = 0$

(D) इनमें से कोई नहीं

Ans. ABC

Sol. $K_1 = K_2 = K_3 = K_4 = 0$

64. If $A(\vec{a}); B(\vec{b}); C(\vec{c})$ and $D(\vec{d})$ are four points such that $\vec{a} = -2\hat{i} + 4\hat{j} + 3\hat{k}$; $\vec{b} = 2\hat{i} - 8\hat{j}$; $\vec{c} = \hat{i} - 3\hat{j} + 5\hat{k}$; $\vec{d} = 4\hat{i} + \hat{j} - 7\hat{k}$, d is the shortest distance between the lines AB and CD, then which of the following is True?

(A) $d = 0$, hence AB and CD intersect

(B) $d = \frac{[\vec{AB} \vec{CD} \vec{BD}]}{|\vec{AB} \times \vec{CD}|}$

(C) AB and CD are skew lines and $d = \frac{23}{13}$

(D) $d = \frac{[\vec{AB} \vec{CD} \vec{AC}]}{|\vec{AB} \times \vec{CD}|}$

यदि $A(\vec{a}); B(\vec{b}); C(\vec{c})$ तथा $D(\vec{d})$ चार बिन्दु इस प्रकार है कि $\vec{a} = -2\hat{i} + 4\hat{j} + 3\hat{k}$; $\vec{b} = 2\hat{i} - 8\hat{j}$; $\vec{c} = \hat{i} - 3\hat{j} + 5\hat{k}$; $\vec{d} = 4\hat{i} + \hat{j} - 7\hat{k}$, d रेखाओं AB तथा CD के मध्य न्यूनतम दूरी है, तो निम्न में से कौनसा कथन सत्य है?

(A) $d = 0$ तथा AB तथा CD प्रतिच्छेद करती है

(B) $d = \frac{[\vec{AB} \vec{CD} \vec{BD}]}{|\vec{AB} \times \vec{CD}|}$

(C) AB तथा CD विसमतलीय रेखाएँ है तथा $d = \frac{23}{13}$

(D) $d = \frac{[\vec{AB} \vec{CD} \vec{AC}]}{|\vec{AB} \times \vec{CD}|}$

Ans. BCD

Sol. $\vec{AB} = \vec{b} - \vec{a} = 4\hat{i} - 12\hat{j} - 3\hat{k}$

$\vec{CD} = \vec{d} - \vec{c} = 3\hat{i} + 4\hat{j} - 12\hat{k}$

$\vec{AC} = \vec{c} - \vec{a} = 3\hat{i} - 7\hat{j} + 2\hat{k}$

$\vec{BD} = \vec{d} - \vec{b} = 2\hat{i} + 9\hat{j} - 7\hat{k}$

By definition $d = \frac{(\vec{AB} \times \vec{CD}) \cdot \vec{AC}}{|\vec{AB} \times \vec{CD}|}$... (i)

$= \frac{(\vec{AB} \times \vec{CD}) \cdot \vec{BD}}{|\vec{AB} \times \vec{CD}|}$... (ii)

$\vec{AB} \times \vec{CD} = 13(12\hat{i} + 3\hat{j} + 4\hat{k}) \quad \therefore \quad |\vec{AB} \times \vec{CD}| = 169$

$$\therefore d = \frac{13(12\hat{i} + 3\hat{j} + 4\hat{k})}{169} \cdot (3\hat{i} - 7\hat{j} + 2\hat{k}) = \frac{23}{13} \quad \text{using (i)}$$

$$\text{also } d = \frac{13(12\hat{i} + 3\hat{j} + 4\hat{k})}{169} \cdot (2\hat{i} - 9\hat{j} + 7\hat{k}) = \frac{23}{13} \quad \text{using (ii)}$$

65. Given the equations of the line $3x - y + z + 1 = 0$, $5x + y + 3z = 0$. Then which of the following is correct ?

(A) symmetrical form of the equations of line is $\frac{x}{2} = \frac{y - \frac{1}{8}}{-1} = \frac{z + \frac{5}{8}}{1}$

(B) symmetrical form of the equations of line is $\frac{x + \frac{1}{8}}{1} = \frac{y - \frac{5}{8}}{1} = \frac{z}{-2}$

(C) equation of the plane through $(2, 1, 4)$ and perpendicular to the given lines is $2x - y + z - 7 = 0$

(D) equation of the plane through $(2, 1, 4)$ and perpendicular to the given lines is $x + y - 2z + 5 = 0$

एक रेखा का समीकरण $3x - y + z + 1 = 0$, $5x + y + 3z = 0$ है। तो निम्न में से कौन सही है?

(A) रेखा का समित रूप है $\frac{x}{2} = \frac{y - \frac{1}{8}}{-1} = \frac{z + \frac{5}{8}}{1}$

(B) रेखा का समित रूप है $\frac{x + \frac{1}{8}}{1} = \frac{y - \frac{5}{8}}{1} = \frac{z}{-2}$

(C) $(2, 1, 4)$ से गुजरने वाला तथा दिये गये रेखा की लम्बवत् समतल का समीकरण $2x - y + z - 7 = 0$ है

(D) $(2, 1, 4)$ से गुजरने वाला तथा दिये गये रेखा की लम्बवत् समतल का समीकरण $x + y - 2z + 5 = 0$ है

Ans. BD

Sol. Let a, b, c be the d.r.'s. of the given line. Then

we have $3a - b + c = 0$

$$5a + b + 3c = 0$$

Solving we get $\frac{a}{1} = \frac{b}{1} = \frac{c}{-2}$

Again, suppose the given line intersect the plane $z = 0$ at $(x_1, y_1, 0)$ then $3x_1 - y_1 + 1 = 0$ and $5x_1 + y_1 = 0$

Solving we get $x_1 = -\frac{1}{8}, y_1 = \frac{5}{8}$

Hence the symmetrical form of the line is $\frac{x + \frac{1}{8}}{1} = \frac{y - \frac{5}{8}}{1} = \frac{z}{-2}$

Equation of plane through (2, 1, 4) is

$$A(x - 2) + B(y - 1) + c(z - 4) = 0$$

when $A = 1$, $b = 1$ and $c = -2$

$$\Rightarrow x - 2 + y - 1 - 2(z - 4) = 0 \Rightarrow x + y - 2z + 5 = 0$$

66. Given three vectors

$$\vec{U} = 2\hat{i} + 3\hat{j} - 6\hat{k}; \quad \vec{V} = 6\hat{i} + 2\hat{j} + 3\hat{k}; \quad \vec{W} = 3\hat{i} - 6\hat{j} - 2\hat{k}$$

Which of the following hold good for the vectors $\vec{U}$, $\vec{V}$ and $\vec{W}$?

(A) $\vec{U}$, $\vec{V}$, $\vec{W}$ are linearly dependent

(B) $(\vec{U} \times \vec{V}) \times \vec{W} = \vec{0}$

(C) $\vec{U}$, $\vec{V}$ and $\vec{W}$ form a triplet of mutually perpendicular vectors

(D) $\vec{U} \times (\vec{V} \times \vec{W}) = \vec{0}$

दिया गया है कि तीन सदिश

$$\vec{U} = 2\hat{i} + 3\hat{j} - 6\hat{k}; \quad \vec{V} = 6\hat{i} + 2\hat{j} + 3\hat{k}; \quad \vec{W} = 3\hat{i} - 6\hat{j} - 2\hat{k}$$

निम्न में से कौनसे कथन सदिशों $\vec{U}$, $\vec{V}$ तथा $\vec{W}$ के लिए सही है?

(A) $\vec{U}$, $\vec{V}$, $\vec{W}$ रेखिक स्वतंत्र नहीं है

(B) $(\vec{U} \times \vec{V}) \times \vec{W} = \vec{0}$

(C) $\vec{U}$, $\vec{V}$ तथा $\vec{W}$ परस्पर लम्बवत् है

(D) $\vec{U} \times (\vec{V} \times \vec{W}) = \vec{0}$

Ans. BCD

Sol. If may be observed that $[\vec{U} \vec{V} \vec{W}] = \begin{vmatrix} 2 & 3 & -6 \\ 6 & 2 & 3 \\ 3 & -6 & -2 \end{vmatrix} = 343 \neq 0$

$\Rightarrow \vec{U}, \vec{V}, \vec{W}$ are non coplanar hence linearly independent

Further $\vec{U} \times \vec{V} = \vec{W}$ and $\vec{V} \times \vec{W} = \vec{U}$

$\Rightarrow$ They form a right handed triplet of mutually perpendicular vectors and of course!

$$(\vec{U} \times \vec{V}) \times \vec{W} = \vec{0} = \vec{U} \times (\vec{V} \times \vec{W})$$

67. Consider the following four pairs of lines in column-I and match them with one or more entries in column-II :

Column-I

(A) $L_1 : x = 1 + t, y = t, z = 2 - 5t$

$L_2 : \vec{r} = (2, 1, -3) + \lambda(2, 2, -10)$

(B) $L_1 : \frac{x-1}{2} = \frac{y-3}{2} = \frac{z-2}{-1}$

$L_2 : \frac{x-2}{1} = \frac{y-6}{-1} = \frac{z+2}{3}$

(C) $L_1 : x = -6t, y = 1 + 9t, z = -3t$

$L_2 : x = 1 + 2s, y = 4 - 3s, z = s$

(D) $L_1 : \frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$

$L_2 : \frac{x-3}{-4} = \frac{y-2}{-3} = \frac{z-1}{2}$

Column-II

(P) non coplanar lines

(Q) lines lie in a unique plane

(R) infinite planes containing both the lines

(S) lines do not intersect

माना Column-I में चार रेखा युग्म हैं एवं Column-II उनसे सम्बन्धित एक या एक से अधिक प्रविष्टि पंजिकायें हैं :

Column-I

(A) $L_1 : x = 1 + t, y = t, z = 2 - 5t$

$L_2 : \vec{r} = (2, 1, -3) + \lambda(2, 2, -10)$

(B) $L_1 : \frac{x-1}{2} = \frac{y-3}{2} = \frac{z-2}{-1}$

$L_2 : \frac{x-2}{1} = \frac{y-6}{-1} = \frac{z+2}{3}$

(C) $L_1 : x = -6t, y = 1 + 9t, z = -3t$

$L_2 : x = 1 + 2s, y = 4 - 3s, z = s$

(D) $L_1 : \frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$

$L_2 : \frac{x-3}{-4} = \frac{y-2}{-3} = \frac{z-1}{2}$

Column-II

(P) विसमतलीय रेखायें

(Q) अद्वितीय तल पर स्थित रेखायें

(R) दोनों रेखाओं से अनन्त तल गुजरते हैं

(S) रेखा प्रतिच्छेद नहीं करती है

Ans. (A) R (B) Q (C) QS (D) PS

Sol. (A) $L_1 : \frac{x-1}{1} = \frac{y-0}{1} = \frac{z-2}{-5}; \quad \vec{V}_1 = \hat{i} + \hat{j} - 5\hat{k}$

$L_2 : \frac{x-2}{2} = \frac{y-1}{2} = \frac{z+3}{-10}; \quad \vec{V}_2 = 2(\hat{i} + \hat{j} - 5\hat{k})$

Hence lines are parallel and both contains the points (1, 0, 2) and (2, 1, -3) $\Rightarrow$ coincident line both L_1 and L_2 may lie in an infinite number of planes hence $\Rightarrow$ (R)

$$(B) \quad \left. \begin{array}{l} \vec{V}_1 = 2\hat{i} + 2\hat{j} - \hat{k} \\ \vec{V}_2 = \hat{i} - 2\hat{j} + 3\hat{k} \end{array} \right\} \Rightarrow \text{lines not parallel}$$

Also both intersect at (3, 5, 1)

Hence lines are intersecting hence they lie on a unique plane $\Rightarrow$ (P)

$$(C) \quad L_1 : \frac{x-0}{-6} = \frac{y-1}{9} = \frac{z-0}{-3} = t$$

$$L_2 : \frac{x-1}{2} = \frac{y-4}{-3} = \frac{z-0}{1} = s$$

$$\left. \begin{array}{l} L_1 \text{ is parallel to } -6\hat{i} + 9\hat{j} - 3\hat{k} \\ L_2 \text{ is parallel to } 2\hat{i} - 3\hat{j} + \hat{k} \end{array} \right\} \Rightarrow \text{lines parallel but not coincident}$$

as (0, 1, 0) does not lie on L_2 , not intersecting

Hence L_1, L_2 lies in a unique plane $\Rightarrow$ (Q), (S)

(D) Lines are skew can be verified $\Rightarrow$ (P), (S)

68. $P(0, 3, -2)$; $Q(3, 7, -1)$ and $R(1, -3, -1)$ are 3 given points. Let L_1 be the line passing through P and Q and L_2 be the line through R and parallel to the vector $\vec{V} = \hat{i} + \hat{k}$:

Column-I

- (A) perpendicular distance of P from L_2
(B) shortest distance between L_1 and L_2
(C) area of the triangle PQR

(D) distance from (0, 0, 0) to the plane PQR

Column-II

(P) $7\sqrt{3}$

(Q) 2

(R) 6

(S) $\frac{19}{\sqrt{147}}$

$P(0, 3, -2)$; $Q(3, 7, -1)$ तथा $R(1, -3, -1)$ तीन दिये हुये बिन्दु है। यदि L_1 रेखा बिन्दु P तथा Q को मिलाती है तथा L_2 रेखा R से गुजरती है एवं सदिश $\vec{V} = \hat{i} + \hat{k}$ के समानान्तर है :

Column-I

- (A) P की L_2 से लम्बवत् दूरी है
(B) L_1 तथा L_2 के मध्य न्यूनतम दूरी है
(C) ΔPQR का क्षेत्रफल

(D) (0, 0, 0) की PQR समतल से दूरी है

Column-II

(P) $7\sqrt{3}$

(Q) 2

(R) 6

(S) $\frac{19}{\sqrt{147}}$

Ans. (A) R (B) Q (C) P (D) S

Sol. $L_1: \frac{x}{3} = \frac{y-3}{4} = \frac{z+2}{1} \dots(1)$
(passing through P and Q)

$L_2: \frac{x-1}{1} = \frac{y-3}{0} = \frac{z+1}{1} \dots(2)$

(passing through R and parallel to $\vec{v} = \hat{i} + \hat{k}$)

(A) distance of P(0, 3, -2) from L_2

$$\overrightarrow{PN} = (t+1)\hat{i} - 6\hat{j} + 2(t-1)\hat{k}$$

now $\overrightarrow{PN} \cdot \vec{V} = 0 \Rightarrow [(t+1)\hat{i} - 6\hat{j} + (t+1)\hat{k}] \cdot (\hat{i} + \hat{k}) = 0$

$$(t+1) + (t+1) = 0 \Rightarrow t = -1$$

hence $\overrightarrow{PN} = -6\hat{j}$

$$|\overrightarrow{PN}| = |-6\hat{j}| = 6 \text{ Ans.} \Rightarrow (R)$$

Distance between L_1 and L_2

Equation of plane containing L_1 and parallel to L_2

$$Ax + B(y-3) + C(z+2) = 0$$

where $3A + 4B + C = 0$

and $A + 0B + C = 0 \Rightarrow A + C = 0$

$$C = \lambda, A = -\lambda, B = +\lambda/2$$

$\therefore$ equation of plane

$$-\lambda x + \frac{\lambda}{2}(y-3) + \lambda(z+2) = 0$$

$$2x - y + 3 - 2z - 4 = 0$$

$$2x - y - 2z = 1 \dots(1)$$

now distance of the point (1, -3, -1) lying on the line L_2 from the plane (1)

$$d = \left| \frac{2+3+2-1}{3} \right| = 2 \text{ Ans.} \Rightarrow (Q)$$

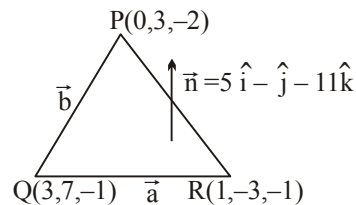
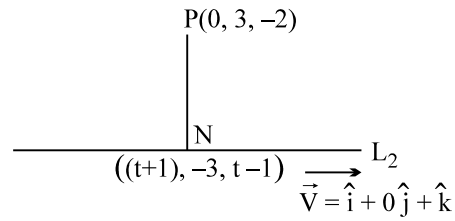
Area of ΔPQR

$$\overrightarrow{QR} = \vec{a} = 2\hat{i} + 10\hat{j} + 0\hat{k}$$

$$\overrightarrow{QP} = \vec{b} = 3\hat{i} + 4\hat{j} + \hat{k}$$

$$\vec{a} \times \vec{b} = 2 \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 10 & 0 \\ 3 & 4 & 1 \end{vmatrix} = 2[\hat{i}(5) - \hat{j}(1) + \hat{k}(4-15)] = 2[5\hat{i} - \hat{j} - 11\hat{k}]$$

$$\frac{|\vec{a} \times \vec{b}|}{2} = \sqrt{25+1+121} = \sqrt{147} = \sqrt{3 \cdot 49} = 7\sqrt{3} \text{ Ans.} \Rightarrow (P)$$



69. Match the following Column :

Column-I

Column-II

(A) Let O be an interior point of ΔABC such that

(P) 0

$\vec{OA} + 2\vec{OB} + 3\vec{OC} = \vec{0}$, then the ratio of the area of ΔABC to the area of ΔAOC , is with O is the origin

(B) Let ABC be a triangle whose centroid is G, orthocentre is H and

(Q) 1

circumcentre is the origin 'O'. If D is any point in the plane of the triangle such that no three of O, A, B, C and D are collinear satisfying

(R) 2

the relation $\vec{AD} + \vec{BD} + \vec{CH} + 3\vec{HG} = \lambda \vec{HD}$

then the value of the scalar ' λ ' is

(C) If $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ are non zero vectors such that no three of them are in the same plane and no two are orthogonal then the value

(S) 3

of the scalar $\frac{(\vec{b} \times \vec{c}) \cdot (\vec{a} \times \vec{d}) + (\vec{c} \times \vec{a}) \cdot (\vec{b} \times \vec{d})}{(\vec{a} \times \vec{b}) \cdot (\vec{d} \times \vec{c})}$ is

निम्न Column को सुमेलित कीजिए :

Column-I

Column-II

(A) माना O त्रिभुज ΔABC के अन्तः में स्थित एक बिन्दु इसप्रकार है कि

(P) 0

$\vec{OA} + 2\vec{OB} + 3\vec{OC} = \vec{0}$ तो ΔABC का क्षेत्रफल का ΔAOC के क्षेत्रफल के साथ अनुपात है, जहाँ O मूल बिन्दु है

(B) माना ABC एक त्रिभुज जिसका केन्द्रक G, लम्बकेन्द्र H तथा परिकेन्द्र

(Q) 1

मूल बिन्दु 'O' है। यदि D एक ऐसा बिन्दु त्रिभुज के तल में इसप्रकार है

कि जिसके लिए O, A, B, C तथा D में से कोई भी तीन समरेखीय नहीं है।

(R) 2

एवं $\vec{AD} + \vec{BD} + \vec{CH} + 3\vec{HG} = \lambda \vec{HD}$ को सन्तुष्ट करते हैं, तो अदिश ' λ '=

(C) यदि $\vec{a}$, $\vec{b}$, $\vec{c}$ तथा $\vec{d}$ चार अशून्य सदिश हैं। इनमें से कोई भी तीन समतलीय

(S) 3

नहीं हैं। तथा कोई भी दो लम्बवत् नहीं हैं, तो अदिश

$\frac{(\vec{b} \times \vec{c}) \cdot (\vec{a} \times \vec{d}) + (\vec{c} \times \vec{a}) \cdot (\vec{b} \times \vec{d})}{(\vec{a} \times \vec{b}) \cdot (\vec{d} \times \vec{c})}$

Ans. (A) S (B) R (C) Q

Sol. (A)
$$\frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta AOC} = \frac{\frac{1}{2} |\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|}{\frac{1}{2} |\vec{a} \times \vec{c}|}$$

now $\vec{a} + 2\vec{b} + 3\vec{c} = \vec{0}$

cross with $\vec{b}$, $\vec{a} \times \vec{b} + 3\vec{c} \times \vec{b} = \vec{0} \Rightarrow \vec{a} \times \vec{b} = 3(\vec{b} \times \vec{c})$

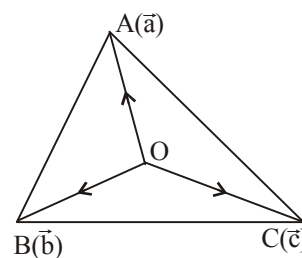
$$\text{cross with } \vec{a}, \quad 2\vec{a} \times \vec{b} + 3\vec{a} \times \vec{c} = 0 \quad \Rightarrow \quad \vec{a} \times \vec{b} = \frac{3}{2}(\vec{c} \times \vec{a})$$

$$\therefore \quad \vec{a} \times \vec{b} = \frac{3}{2}(\vec{c} \times \vec{a}) = 3(\vec{b} \times \vec{c})$$

$$\text{Let } (\vec{c} \times \vec{a}) = \vec{p}$$

$$\vec{a} \times \vec{b} = \frac{3\vec{p}}{2}; \quad \vec{b} \times \vec{c} = \frac{\vec{p}}{2}$$

$$\therefore \quad \text{ratio} = \frac{|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|}{|\vec{c} \times \vec{a}|} = \frac{\left| \frac{3\vec{p}}{2} + \frac{\vec{p}}{2} + \vec{p} \right|}{|\vec{p}|} = \frac{3|\vec{p}|}{|\vec{p}|} = 3 \text{ Ans. } \Rightarrow \quad (S)$$



$$(B) \quad \text{LHS} = \vec{d} - \vec{a} + \vec{d} - \vec{b} + \vec{h} - \vec{c} + 3(\vec{g} - \vec{h})$$

$$= 2\vec{d} - (\vec{a} + \vec{b} + \vec{c}) + 3\frac{(\vec{a} + \vec{b} + \vec{c})}{3} - 2\vec{h}$$

$$= 2\vec{d} - 2\vec{h} = 2(\vec{d} - \vec{h}) = 2\vec{HD} \quad \Rightarrow \quad \lambda = 2 \text{ Ans. } \Rightarrow \quad (R)$$

$$(C) \quad (\vec{b} \times \vec{c}) \cdot (\vec{a} \times \vec{d}) = \begin{vmatrix} \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{d} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{d} \end{vmatrix}$$

$$\text{|||ly compute others which gives (1)} \quad \Rightarrow \quad (Q)$$

70. Consider the following linear equations $ax + by + cz = 0$, $bx + cy + az = 0$, and $cx + ay + bz = 0$
Match the conditions/ expressions in Column I with statements in Column II.

Column I

- (A) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$
(B) $a + b + c = 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$
(C) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$
(D) $a + b + c = 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$

Column II

- (P) the equation represent planes meeting only at a single point
(Q) the equation represent the line $x = y = z$
(R) the equation represent identical planes
(S) the equation represent the whole of the three dimensional space

माना रैखिक समीकरणों $ax + by + cz = 0$, $bx + cy + az = 0$, तथा $cx + ay + bz = 0$ है।

Column-I में दिये गये प्रतिबन्धों/व्यंजकों का Column-II में दिये गये सुमेल कीजिये :

Column I

- (A) $a + b + c \neq 0$ तथा $a^2 + b^2 + c^2 = ab + bc + ca$
(B) $a + b + c = 0$ तथा $a^2 + b^2 + c^2 \neq ab + bc + ca$
(C) $a + b + c \neq 0$ क $a^2 + b^2 + c^2 \neq ab + bc + ca$

Column II

- (P) समीकरणों समतलों को प्रदर्शित करती है जो केवल एक बिन्दु पर मिलते हैं।
(Q) समीकरणों रेखा $x = y = z$ को प्रदर्शित करती है
(R) समीकरणों समान तलों को प्रदर्शित करती है

(D) $a + b + c = 0$ तथा

$$a^2 + b^2 + c^2 = ab + bc + ca$$

(S) समीकरणें सम्पूर्ण त्रिविमीय समष्टि को प्रदर्शित करती है

Ans. (A) R (B) Q (C) P (D) S

Sol. Let $\Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$

$$= \frac{-1}{2} (a + b + c) [(a - b)^2 + (b - c)^2 + (c - a)^2]$$

(A) If $a + b + c \neq 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$

$$\Rightarrow (a - b)^2 + (b - c)^2 + (c - a)^2 = 0$$

$$\Rightarrow \Delta = 0 \text{ and } a = b = c \neq 0$$

$\Rightarrow$ the equation represent identical planes.

(B) $a + b + c = 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$

$$\Rightarrow \Delta = 0$$

Since all the three planes pass through (1,1,1)

So equation of the line of intersection of these

plane will be $\frac{x-0}{1-0} = \frac{y-0}{1-0} = \frac{z-0}{1-0}$

(C) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$

$$\Rightarrow \Delta \neq 0$$

$\Rightarrow$ the equations represent planes meeting at only one point i.e. (0,0,0)

(D) $a + b + c = 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$

$$\Rightarrow a = b = c = 0$$

$\Rightarrow$ the equations represent whole of the three dimensional space.